Automated Metal Particulate Counting for Handpiece Manufacturing

Authors: Matthew Luong, Yolanda Gomez, Isaac Hoyt, Brian Ngo, Landon Stein, Tandol Mathew Jerthsattayanukul

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Executive Summary

Alcon's manufacturing process of the Active Sentry handpieces undergo a strict quality control test where technicians evaluate the handpieces for any titanium residue during their machining process. The handpieces are flushed with water on a mesh filter in which highly-trained technicians examine the filter under a microscope for any metal particulate. Alcon's current method with the technicians is a time consuming process taking around 18-30 minutes to complete an inspection of one batch of 10 handpieces. The Alcon's production of the Active Sentry handpieces requires a specialized technician, which takes months to train and requires lots of resources. As said earlier, one technician takes around 18-30 minutes per one batch to fully inspect one handpiece creating a bottleneck in Alcon's production line as the handpiece is a part of the Centurion Vision system that Alcon is selling to the public. The device solves the problem of Alcon's current method to inspect the handpieces for any metal particulate by automating the process, thus reducing the intensive labor demands the technicians go through. The software analyzes pictures of the filter taken from a microscope and reports the results that comply with Alcon's standards. The device is effective and most importantly, replicable. Alcon can set up a station of multiple devices running at the same time, increasing the productivity of the handpieces compared to Alcon's current method. The device consists of a motorized stage controller that technicians slide the filter cartridge through. The stage controller, under the microscope, covers all areas of the filter while the software of the system is taking pictures of the filter. The software then analyzes the pictures for any metal particulate matter and returns a report that is based on Alcon's criteria that determines if the handpieces are safe for surgical use. One technician can utilize multiple devices at once, which is the target audience for the device. In addition, the labor required to use the devices is significantly less if they were to inspect the filters themselves. The project is closely related to economic factors as the device is aimed to streamline processes and increase productivity by automating repetitive tasks, freeing up human resources for more strategic work.

Introduction

Alcon Inc's current method for conducting their Metal Particulate Count Test is inefficient and labor-intensive. On their manufacturing line, they conduct a test to ensure that residual titanium particles from the machining process are within the acceptable limits and maintain patient safety.

Their manual inspection under a microscope, as performed by trained technicians, can take fifteen minutes per filter and is a production bottleneck. In a highly controlled medical device environment, this time-intensive process slows down manufacturing output, increases operational costs, and limits scalability. Moreover, training skilled technicians is resource-intensive – exacerbating inefficiencies.

The current standard involves manually inspecting all areas of a membrane filter underneath a digital microscope, where a technician visually identifies and counts the titanium particles. The process is subjective, time consuming, and prone to variability. As of right now, no comprehensive automation or image analysis software is being used in Alcon's workflow for this essential quality assurance procedure. The Active Sentry handpiece is a component of Alcon's Centurion eye surgery system, and features a housing that is machined from titanium. Titanium particles leftover from the machining process can irritate the delicate tissues of the retina during a procedure, leading to tears or even retinal detachment. Therefore, it is necessary to comply with standards such as USP 788, which defines allowable particle limits, and ASTM F3127-22, which validates cleaning processes used in medical device manufacturing.

After assembly, the handpieces undergo a testing procedure that concludes with a fluid flush of the device. Residual particles are suspended in the fluid and then captured by a membrane filter. Currently, a technician manually inspects the filter with a microscope to count and measure the size of any titanium particles that they find. The nature of manually counting the particles is inherently inefficient and unsustainable for high-volume production. Again, this reliance on human labor demands training, introduces variability, and limits scalability. An automated, more consistent, system is necessary to meet the demands of production while also maintaining the safety standards that ophthalmic surgery requires.

The Alcount device integrates a motorized X-Y stage controller, digital microscopy, and a convolutional neural network (YOLOv5) to automate particle detection and classification. This system allows a technician to run multiple instances of the test in parallel, significantly reducing the labor time and improving consistency. Particle analysis software does currently exist in other industries (e.g. contamination testing in semiconductor manufacturing), but few off-the-shelf solutions are tailored to particle detection in medical device QA applications. The design is novel in its integration and application to Alcon's process – reducing the amount of required training and increasing scalability.

It is a cost-effective, replicable improvement for precision QA processes in medical device manufacturing. Compared to the standard process, Alcount reduces inspection time without compromising inspection quality; furthermore, by lowering inspection time, handpieces will be cleared faster for downstream production – lowering costs for Alcon while increasing manufacturing and economic output.

To improve the standard process, the custom automated particle analysis platform combines mechanical and software systems. A filter membrane is placed on a motorized stage controlled through Arduino where a microscope will capture images as the stage moves the filter underneath the camera. From there, the images are analyzed in real-time using python software that leverages YOLOv5 for particle detection. The software quantitatively measures particle count and size, compares those results with Alcon's acceptance thresholds, and generates a pass/fail report. This solution transforms the manual inspection into an automated workflow.

Initial Design Phase and Design Controls Regulation

The overall design requirements can be summarized into this table. The overall design requirements prioritize autonomy, efficiency, accuracy, and replicability. The design needs to be autonomous and time-efficient especially as that is the project's main focus and scope. The overall design requirements also need to be inline with Alcon's current testing standards, and technicians need to be satisfied with the new product they will be using.

Autonomous	Needs to identify and analyze metal particulate

	autonomously Pass/Fail
Time-Efficient	Needs to be faster than the current method Time < 18-30 minutes per quantity 10
Cost-Efficient	Needs to be an affordable device for the company to reproduce Cost < $2000 (Senior Design Budget)
Identifies Particle Count	Quantifies the particulate count per given area and measures the size of each individual particle Pass/Fail
Determines Safe for Use	Needs to determine if the handpiece is ready for production given Alcon's testing criteria Pass/Fail
Records Necessary Information	Needs the documentation of each handpiece once it has passed the test Pass/Fail
Integration of AI	Machine learning software can improve the accuracy of the script we are using to analyze particulate Pass/Fail
Useability	How accessible and easy-to-use is the prototype to technicians Pass/Fail

Table 1: System Design Requirements

The design was then divided into a hardware and software design criteria. For the overall mechanical design, it can be summarized into this table. The mechanical design requirements prioritize precision, automation, speed, and cost-efficiency. The design needs to allow the filter to scan the entire area under a microscope in a time that is in accordance with the overall design requirements.

Precision and Accuracy	Ability to accurately pinpoint and survey the entire particulate filter
Automation Capability	Ability to reduce manual work for analyzing the particulate filter
Cost Effectiveness	Total cost of the method with 5 being the most

	cost effective strategy
Speed of Operation	Time it takes to analyze the particulate filter
User-Friendly Interface	How easy and accessible it is to use the method
Maintenance Requirements	How easy and efficient to maintain the system

Table 2: Overall Mechanical Design Requirements

The overall design requirements are also in a 1-5 point scale with 5 being the most proficient and 1 being the least proficient because a couple of designs will be vetted with the table above to determine the best design. With the overall design requirements in mind, some initial design drawings can be seen below.

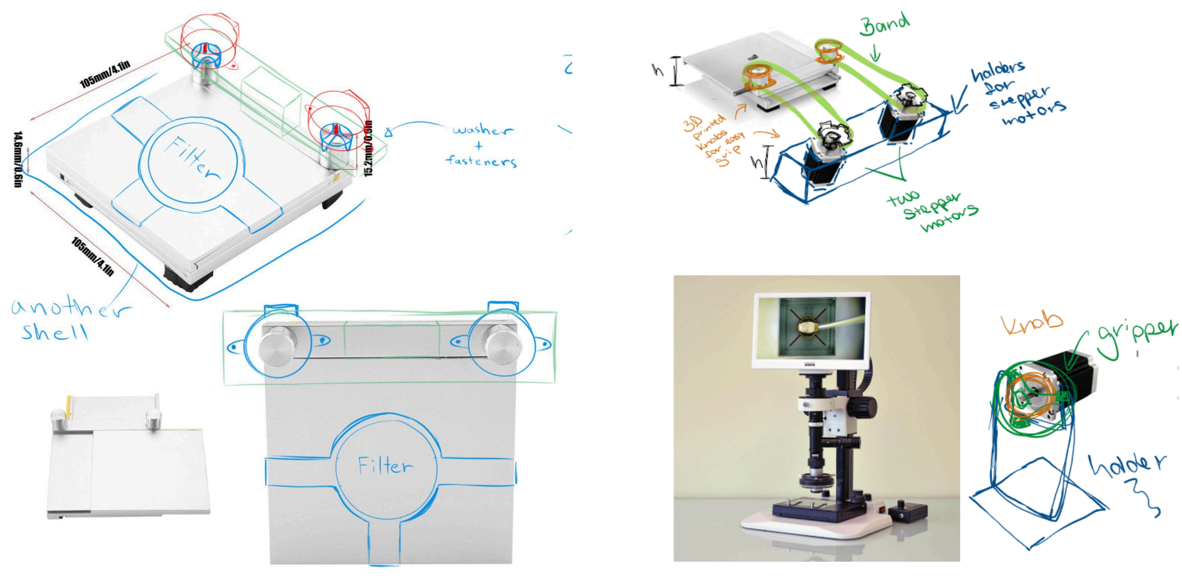


Figure 1: Knob Controlled Stage Controller (Left) and Pulley System Stage Controller (Right)

The idea that was proposed was to modify an existing mechanical stage controller with 3D-printed parts and motorized controllers to make it autonomous. For both of these designs, they followed that idea philosophy. The Knob Controlled Stage Controller (KCSC) consists of a motor holder that will house two stepper motors that will directly control the knobs allowing for X-Y stage movement. In theory, this will cover the entire area of the filter when the technician places the filter in its designated slot. Problems with this design is that the knobs from the mechanical stage controller were too gummy and had too much play, which would limit the accuracy and precision of the stage controller's movement. In addition, the top place where the technician would place the filter is too thin for a shell to be installed. The Pulley System State Controller consists of a conveyor band that is controlled by stepper motors at a short distance away from the stage controller. However, the design also had problems with movement inaccuracies due to band movement, being not easy to replicate, and too time consuming given the time frame for this project.

The ideas were vetted against each other and the current method in a Pugh matrix. With the two proposed designs and current solution, the KCSC had the most potential out of the three, being highly replicable, cost-efficient in terms of purchasing the required components, and the quickest in operation when estimating the approximate time to inspect the entire filter.

Criteria	Baseline (Technician)	Knob Controlled Stage Controller	Pulley System Stage Controller
Precision and Accuracy	5	4	3
Automation Compatibility	1	5	4
Cost Effectiveness	1	4	4
Speed of Operation	2	4	3
User-Friendly Interface	2	3	3
Maintenance Requirements	1	2	2
Total	12	22	19

Table 3: Pugh Matrix of Proposed Designs and Current Method

The first iteration of the KCSC can be seen below with a table of what each part is. The first generation of the design consists of the three originally proposed parts with two new additions being the rails and bracket.

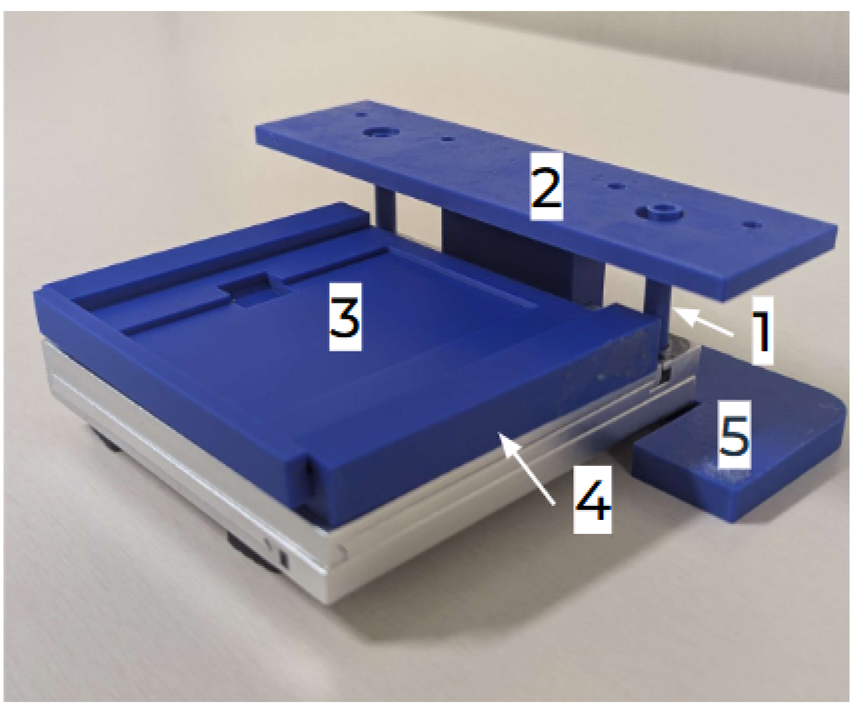


Figure 2: First Generation Knob Controlled Stage Controller

Part #	Name
1	Knobs
2	Motor Holder
3	Stage Top
4	Rails
5	Bracket

Table 4: Labeled Sections of the Knob Controlled Stage Controller

In the original drawing, the gripper knobs were supposed to attach to the knobs that came with the mechanical stage controller. As stated earlier, that would cause too much play, so the solution to work around that was to remove the knobs entirely and replace them. By having a direct connection for the stepper motor and the gears of the stage controller, the play was heavily reduced allowing for more precise movement. The motor holder remains mostly unchanged to the original drawing as it is designed to hold the motors steadily as the state controller moves and does not interfere with the microscope's view.

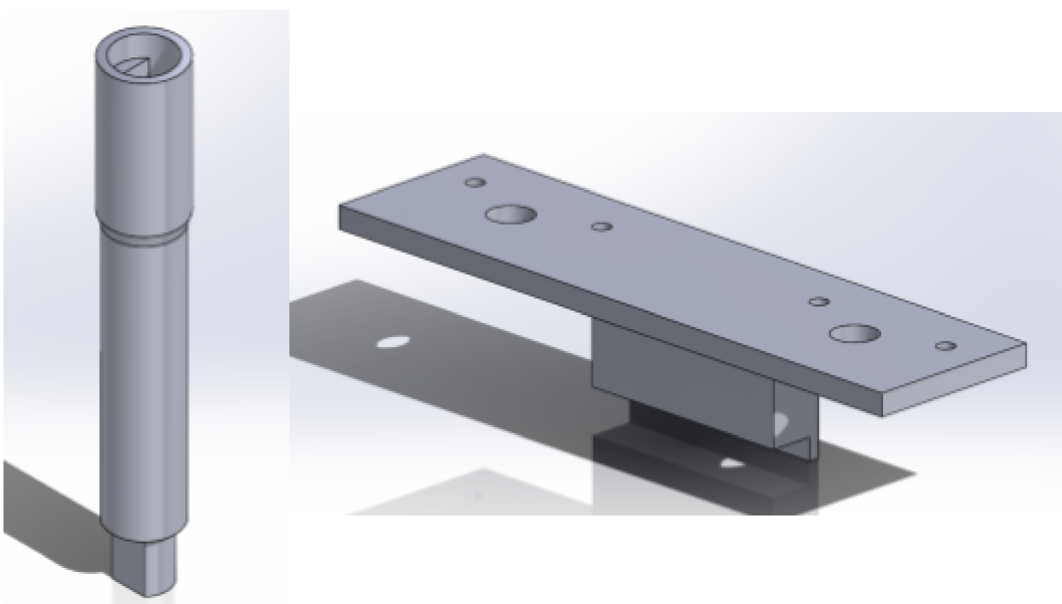


Figure 3: CAD Models of Knob (Left) and Motor Holder (Right)

The shell in the original drawing was replaced with a stage top because the technicians use a filter cartridge to analyze metal particulate under the microscope. The theory behind the design was to allow easy removal of the cartridge after the stage controller had done its job. The rails would be permanently attached to the stage controller and the stage top can be slid out to remove the filter and put in a new one.

The bracket is modeled to the stage controller's dimensions and is adhered under the microscope to ensure repeated placement in the same position for iterative testing. Setting up such parameters reduces the amount of factors that can impede the accuracy and precision of the stage controller's movement.

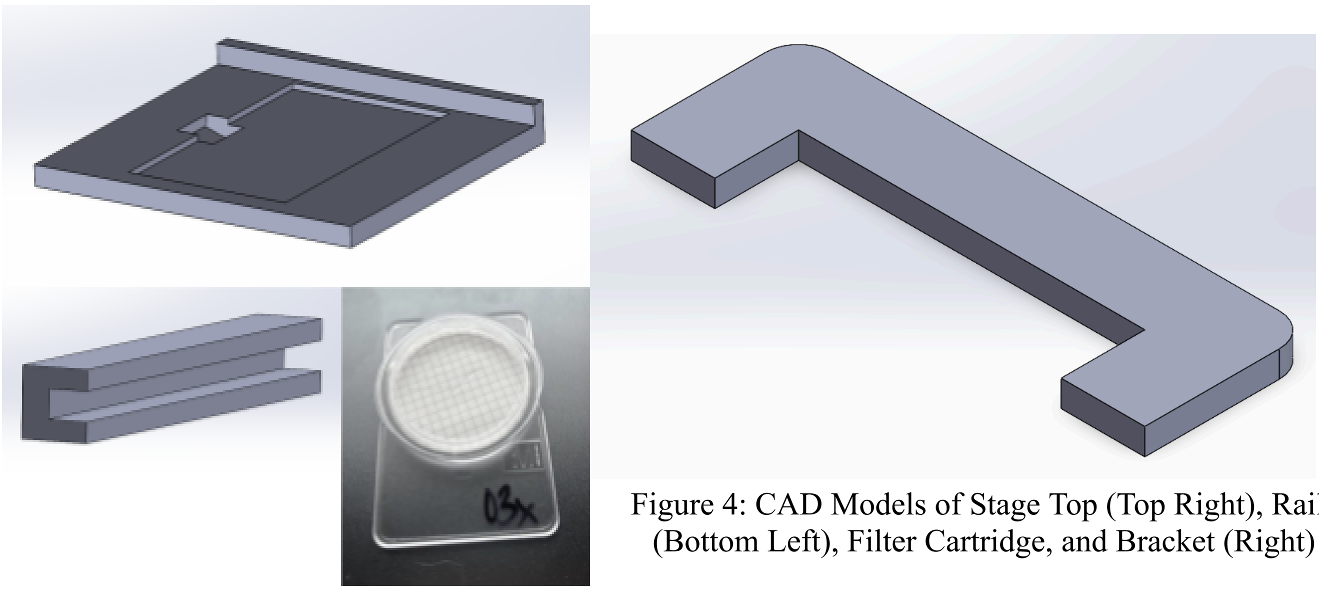


Figure 4: CAD Models of Stage Top (Top Right), Rails (Bottom Left), Filter Cartridge, and Bracket (Right)

For the overall software requirements, they are summarized in a table below similar to the overall hardware requirements. The theory for the software aims to automate particle detection and size measurement with accuracy comparable to a technician, while minimizing human involvement. It must reliably detect titanium particles while rejecting non-titanium ones, integrate seamlessly with existing hardware, and analyze images at least as fast as a technician. Additionally, it should store relevant particle data (ID, location, size), provide a user-friendly interface, and generate reports for the technician to input in their systems.

Particle detection automation	Minimise technician involvement
Particle detection accuracy	Must accurately reject non-titanium particles
Particle size measurement automation	Minimise technician involvement
Particle size measurement accuracy	Must be equivalent to technician measurement accuracy
Integration with existing hardware	Must be plug-and-play, should not unnecessarily increase amount of hardware needed for test
Image processing speed	Should be faster than or equal to technician
Store and track particle data	ID number, centroid location, size measurements
User-friendly interface	Generates report for technician

Table 5: Overall Software Requirements

A house of quality matrix is shown below as the customer requirements (technicians) are vetted against the functions of the proposed software function. The first function of the software is to gather images and get them prepared for the image analysis portion. The second function is to utilize an image analysis software to detect each image for any metal particulate. For the image analysis, an open source tool called YOLOv5 will be used to aid with the accuracy of the metal particulate detection by training it with test images to more accurately spot what is a titanium particle and reject any other objects. After it detects the particle, the software should give out the measurement, categorize it, and return a report to the user after all the images have been analyzed. With the house of quality matrix, developing the object detection is the most important function the software needs followed by the image gathering and particle measurement.

	Desired direction of improvement (↑,0,↓)	↑	↑	↑	↑	↑
1: low, 5: high Customer importance rating	Functional Requirements (How's) → Customer Requirements - (What's) ↓	Image input handling and preprocessing	YOLOv5 object detection	Particle measurement	Particle data storage and tracking	Report generation
5	Particle identification accuracy	1	9			
5	Image processing speed	9	9	9	3	
3	User-friendly interface	9	1	3	3	9
5	Automated particle identification	1	9			
5	Automated particle size measurement	1		9		
3	Generate report	9	3	3	9	9
3	Integration with existing hardware	9	1	1		3
5	Particle size measurement accuracy	1	1	9	3	
5	Store and track particle data (position, size, count)	3	1	3	9	
5	Accurately rejects non-titanium particles	9	9			
	Technical importance score	191	200	156	66	63
	Importance %	25%	26%	21%	9%	8%
	Priorities rank	**2**	**1**	**3**	**4**	**5**

Table 6: House of Quality for Software

With the hardware and software requirements described, our proposed design consists of a motorized stage controller that the user will place a filter cartridge in. The hardware components will aid the software commands to ensure precise functionality. The hardware components consist of 3D-printed mounts to secure the filter membrane and its casing, as well as housings for the stepper motors and knobs designed to fit both the purchased microscope stage and the motors. The software will capture and analyze images of the filter membrane to detect metal particulates, with stepper motors controlling movement. Once the measurements are recorded, the images will be deleted, and a report will be generated for Pass/Fail classification based on a predetermined threshold.

The design addresses the problem by automating the metal particulate test allowing for greater efficiency, decreased labor costs, and multiple stations to be run at once. Alcon will be able to set up multiple iterations of the system increasing the productivity of the handpieces compared to Alcon's current method, which is the technicians. The system is designed to be cheap to reproduce and compact measuring approximately 105 mm x 110 mm and costing less than $100 to produce on the current market.

The device does not require a 30 day review from FDA because the objective of the project aims to improve the test method of the manufacturing of the handpiece. The handpiece itself has already been reviewed as it is a Class 2 device and does not require additional assessment from the FDA.

The engineering standards listed below are the basis for the metal particulate test and a guide to the principles used in the design of the procedure:

ISO 16269-6:2005; Statistical interpretation of data. Determination of statistical tolerance intervals

The standard describes the procedures for establishing tolerance intervals that include a specified proportion of a population with a specified confidence level. It provides methods for both normal distributions and non-normal continuous distributions. This statistical standard is the basis for the determination of error due to providing tolerance intervals for unknown distributions.

ASTM F2459-05; Standard Test Method for Extracting Residue from Metallic Medical Components and Quantifying via Gravimetric Analysis

The standard describes a test method for extracting and quantifying metallic components in medical devices, which is crucial for assessing cleanliness in medical device manufacturing. This standard provides a basis for the criteria regarding metal particulate control.

ISO 13485:2016; Medical devices - Quality management systems - Requirements for regulatory purposes

This international standard specifies the quality management system (QMS) requirements for organizations involved in the design, production, installation, and servicing of medical devices and related services. This ensures organizations meet global regulatory requirements for medical devices.

Department of Health and Human Services Food and Drug Administration 21 CFR Parts 4 and 820; Medical Devices: Quality System Regulation

This regulation describes the current good manufacturing processes (CGMP) requirements of the Quality System (QS). To comply with this regulation, medical device manufacturers are required to establish and maintain a quality system that is appropriate for the medical devices designed or manufactured.

USP 788; Testing for Particles in Injectable Products

This standard describes two test methods suitable for assessing particulate matter present in injectable solutions. Although the AMPC tests for particles present in Alcon’s surgical handpieces are not injectable solutions, it aligns with the microscope assay method outlined in USP 788.

IEC 60601 Medical electronics safety

This is an international standard that specifies the safety and essential performance requirements for medical electrical equipment. It sets out technical requirements for the design, construction, and testing of medical electrical equipment to prevent hazards such as electric shock, fire, and mechanical damage, which will be useful in risk management and writing guidelines for setting up the system.

ISO 14971 Risk management of medical devices

This is the international standard for risk management in medical devices. It provides a structured process for identifying, assessing, and controlling risks associated with medical devices throughout their lifecycle. This process ensures that the benefits of using a medical device outweigh the potential risks, which will be highlighted when assessing potential modes of failure for the device.

The estimated project timeline shown below highlights the key milestones that should be accomplished. During the Fall quarter, conducting research and brainstorming ideas for the design was the best move while also making early attempts to work on the first generation prototype if possible. During the Winter quarter, the prototyping stage will continue while also receiving feedback from mentors with the Spring quarter being the time to test the system and make adjustments accordingly.

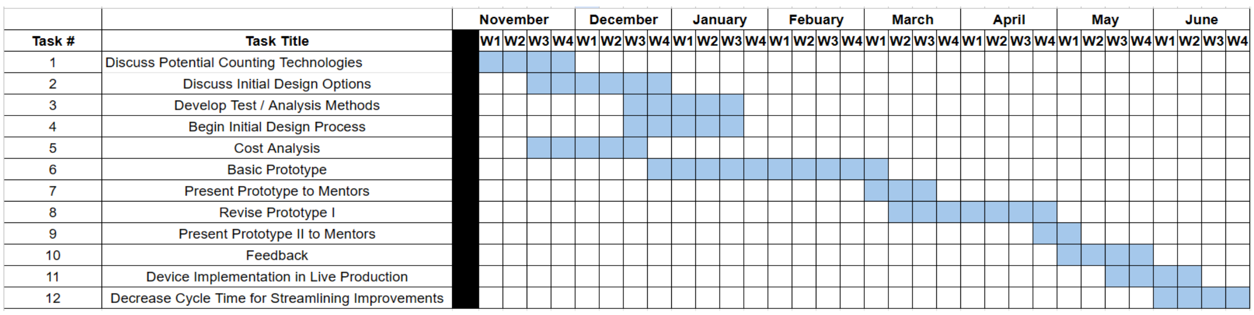


Figure 5: Gantt Chart of Estimated Project Timeline

The budget for this project consists of purchasing components for the stage controller and its circuitry. As for the 3D-printed components, the team has resources to print the parts without needing to purchase a 3D printer. For the software side, a compact microscope with an attached camera is needed to test the image acquisition and train the image analysis software. The estimated cost of all the items is less than the given budget. It is important to note that the microscope Alcon will be using will not be the one that the team is purchasing. The total cost for the estimated budget is approximately $1,189.14 not including tax and shipping costs.

Components	Description	Supplier	Unit Price	Quantity	Total Cost
X-Y Stage	Allows movement to cover the entire filter membrane	Amazon	$45.00	1	$45.00
Stepper Motors	Allows movement of knobs to displace stage	Amazon	$8.99	2	$17.98

Stepper Motor Modules	For precise positioning and movement control	Amazon	$10.28	5 pcs/Set 1 Set	$10.28
Arduino Mega 2560	Microcontroller for controlling stage controller movement	Amazon	$50.94	1	$50.94
Power Supply Cable	Supplies microcontroller with energy	Amazon	$11.99	1	$11.99
Breadboard Wires	Wiring components	Amazon	$5.97	1	$5.97
Nuts and Bolts Assortment Kit	Attaching stepper motor to motor holder	Amazon	$8.99	1	$8.99
Microscope with Camera	Tests for image acquisition and trains image analysis software	Microscope Central	$1,073.99	1	$1,073.99

Table 7: Proposed Budget

Project Team

For this project, the team decided to split the group into two respective groups: one hardware team and one software team. In these subgroups, the team decided on more specific roles within those subgroups.

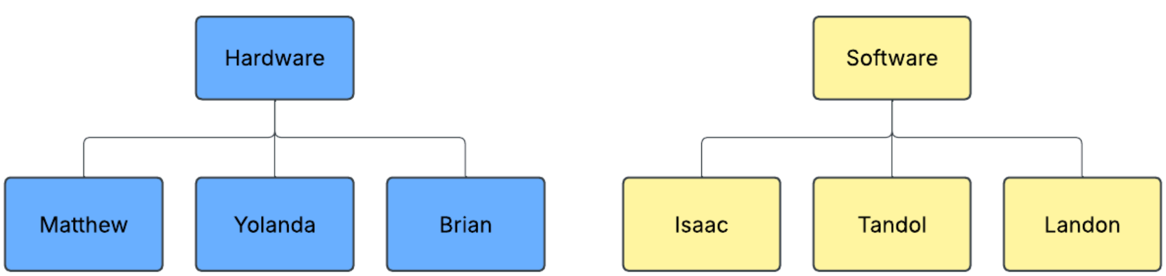


Figure 6: Hardware and Software Teams

Name	Role
Matthew Luong	Hardware Lead
Yolanda Gomez	Manufacturing Lead
Tandol Mathew Jerthsattayanukul	Software Integration Lead
Landon Stein	Software/Hardware Integration
Isaac Hoyt	Team Leader
Brian Ngo	Hardware Integration/Business Advisor

Table 8: Team Member and Respective Roles

Individual Team Member Contributions:

1. Matthew Luong

As the Hardware Lead, he was assigned to design and 3D print components of the automated stage controller. The subtasks in progressing order are as follows: initial design, solid modeling, materials selection, and manufacturing of 3D-printed components.

The initial design process took the hardware design requirements into consideration as the theory behind the design was to produce a stage controller that moves autonomously, accurately, and is time efficient. When designing the 3D parts on Solidworks, the parts were made to be easy for the technicians to use and assemble because usability was a design consideration. When selecting materials for the 3D-printed components, he found the most cost-effective filament because that was an overall design requirement. When manufacturing the 3D-printed components, finding a printer that has precise tolerances was considered because a cleaner part required less maintenance.

Because he was assigned to design and 3D print components of the automated stage controller, the technician is able to place a filter in the stage controller and it will be able to be scanned by the microscope. The automated stage controller is a key component of our system because it is what is going to allow the microscope to receive the necessary information to analyze the filter for metal particulate.

2. Yolanda Gomez

As the Manufacturing Lead, Yolanda Gomez was responsible for producing the physical components of the prototype and ensuring the design was practical, manufacturable, and met performance standards. In addition to fabrication, she played a key role in design optimization by identifying issues that arose during initial usability tests. Her contributions helped improve parts that, while theoretically functional, required refinements for real-world application. These changes ensured that the prototype could be assembled efficiently and operate as intended.

Yolanda's role included subtasks such as modifying CAD models in SolidWorks, operating and post-processing 3D prints, selecting appropriate materials, and documenting each step of the manufacturing process. All tasks were shaped by design constraints like compact size (to fit within a microscope stage), precision tolerances, and low-friction movement. These requirements guided decisions such as print resolution, part orientation, and material properties. To validate the manufactured components, measurements were taken with calipers to confirm dimensional accuracy, and parts were fit-tested with the motion system to ensure proper alignment and mechanical integration.

The components Yolanda produced were critical to the functionality and validation of the system. They supported smooth stage movement, consistent slide positioning, and alignment for accurate particle detection. Validation tests were conducted by integrating the manufactured parts into full system trials and confirming that the prototype met key performance metrics, such as maintaining slide positioning accuracy and supporting a processing speed under 4 minutes per batch. These tests confirmed that her manufacturing work met all necessary criteria and directly supported the successful validation of the overall design.

3. Tandol Mathew Jerthsattayanukul

As the software-integration lead, Tandol was responsible for developing an Arduino firmware and python scripts. The Arduino code contains all the function codes to move the stage control as designed. The python script contains all code that involves moving sequences of the stage controller, image capturing through microscope, YOLO integration for particle detection, and OpenCV to generate particle measurements and draw bounding boxes. His contribution is the keystone that binds the stage controller, imaging hardware, and AI detection into a single automated system.

In addition to developing and integrating the control and analysis code, he is also responsible for prototype testing. After each code is updated, a test is being run on the hardware to see if it is fully optimized, ensuring that the code runs without any hiccups, generates no errors and malfunctions. After each testing is done, he integrates new features into the python script and runs the test again until the prototype is refined.

Since this project is to develop an automated system to facilitate the testing process, his prototype validation was critical to the project, making sure the prototype is well refined and fully functioning.

4. Landon Stein

As the coordinator between the software and hardware teams, Landon was responsible for integrating the multiple subsystems within the overall assembly. He oversaw the designs of the 3D components, providing feedback and suggestions. Further, he set up the preliminary circuit for the motor's controls and its interface with the computer program. He also helped develop the overall assembly.

The design requirements for the circuit were generated in the initial development process. The power draw of the motors, the pin inputs to be used, and the overall circuit design. Each part is integral to the overall assembly of the device. The tasks performed created some of the criteria used in the final design.

5. Isaac Hoyt

As the general team leader and the software team lead, Isaac was responsible for selecting and training the YOLO particle recognition model. Isaac proposed the initial design concept of a moving stage controller in conjunction with a software module for automatically counting particles. Isaac proposed using an open-source YOLO object recognition model to identify particles. He drew up the House of Quality to refine software requirements and determine software modules. He also gathered the training data and trained the YOLO particle recognition model from scratch.

6. Brian Ngo

Brian has been organizing team meetings, creating agendas for the meetings, action plans for each team member on preparing to meet with Alcon mentors, delegating tasks for team members for the homework, creating Gantt charts, and occasionally providing research assistance for the team. Brian also played a role on the hardware side when the prototyping stage occurred, and software development of a user interface at the end.

Detailed Design Phase

The final design was presented after initial prototype testing. The initial testing showed multiple flaws in the original design that needed to be adjusted.

The device is based on an off-the-shelf manual stage controller (stage controller) with separate X-Y axis controls (**Figure 1**). The stage has been modified to be automatically controlled by an Arduino Mega 2560 (Arduino) (**Figure 2**). Two 28BYJ48 stepper motors (motors) are used to move the stage and are connected to the Arduino (**Figure 3, 1**). Between the Arduino and the motors, a ULN2003 provides intermediate control of the motors (**Figure 3, 2).** Custom 3D printed fixtures (printed components) are used to attach the motors to the stage – they are composed of both PLA and Nylon. Lastly, the device comes with a zeroing clamp that allows the stage to be set to its base position (**Figure 4**).

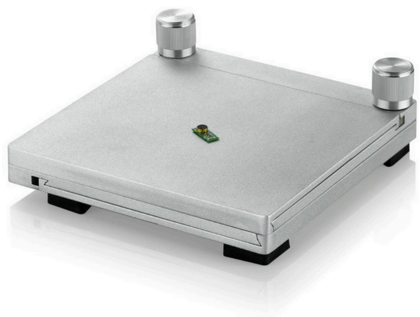


Figure 1: Stage Controller

Figure 2: Arduino Mega 2560

Figure 3 (Top to Bottom): 28BYJ48 Stepper Motor (1), ULN2003 Drive Module (2)

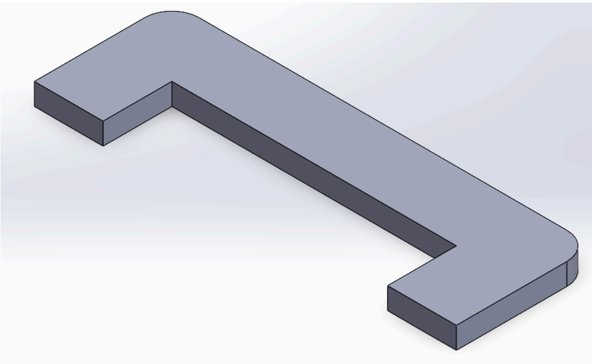


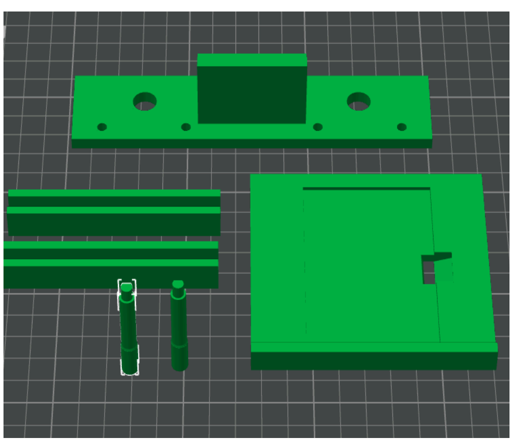
Figure 4: Stage Zeroing Fixture

The printed components are made up of the stage top (**Figure 5, 3**), the motor holder (**Figure 5, 2**), and the gripper knobs (**Figure 5, 1**).

The software initially functioned as intended, however it was missing its object detection from YOLOv4. The software uses Arduino and Python code to move the stage and collect images. The software, combined with the hardware components, moves the entire filter slide under the microscope as intended.

Originally, YOLOv4 was used in the object detection of the device. However, due to compatibility issues, YOLOv5 was used. The initial training data and exclusion criteria pointed out multiple false positives, wherefore new parameters were set and detection became more accurate.

Initially, the hardware design of the device included extra 3D-printed parts – a pair of frame rails (**Figure 4, 2**). These components were in the original design based on preliminary criteria. Further, the printed parts had their dimensions, tolerances, and materials adjusted. Initially, all of the printed components were printed out of PLA. However, after suggestions from Alcon, the material of the gripper knobs was changed to Nylon 12 due to its higher tolerance in the printing process.



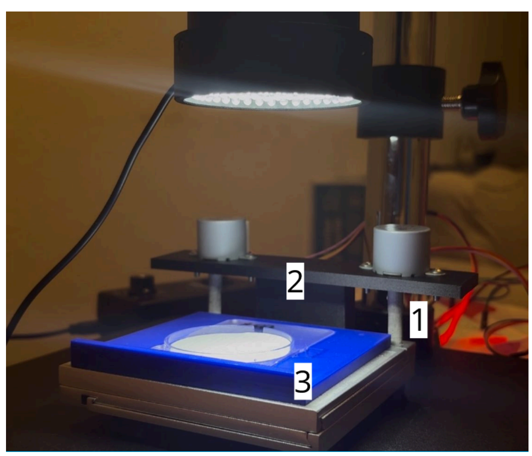


Figure 4, Original 3D Printed Components.
Top to bottom, left to right: Motor Holder (1), Frame Rails (2), Stage Top (3), Gripper Knobs (4).

Figure 5, Final 3D Printed Components. Gripper Knobs (1), Motor Holder (2), Stage Top (3).

In testing, the motors appeared to have reliability issues – new motors were considered but, after further software adjustments, no changes were necessary and the original motors were used between the initial prototype and the final device.

Everything else in the final design remained the same as in the original design – the circuitry and the controllers.

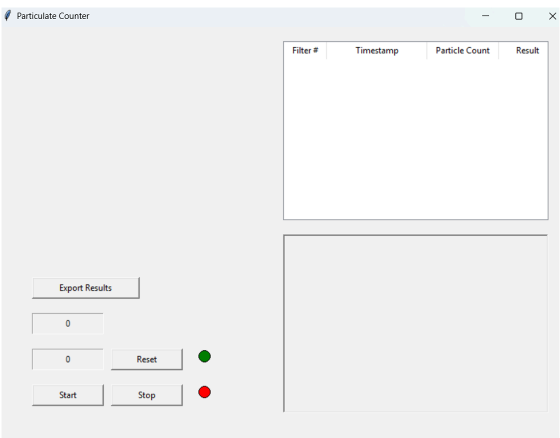


Figure 6: Software Interface

A software interface is included in the final version of the device. Compared to the initial prototype, the user no longer needs to run the code directly and can start the process with one button push. The interface includes a stop/start button, progress and error indicators, filter counter with reset, sample image display, particle count output, session log table, and a pass/fail indicator.

The design of the device allows it to be integrated seamlessly into Alcon's manufacturing line. Alcon's manufacturing line includes microscopes and computers that have software for the image capture of metal particulates. The entirety of the device fits underneath Alcon's microscopes and the code needed to run the device can be easily run on their computers. The device takes minimal extra space in order to function.

Data Sheets:
ULN2003: https://www.ti.com/product/ULN2003A
28BYJ48: https://www.mouser.com/datasheet/2/758/stepd-01-data-sheet-1143075.pdf
Arduino Mega 2560: https://docs.arduino.cc/resources/datasheets/A000067-datasheet.pdf

The device requires an external power source, provided by a 12V power supply.

The total budget used over the course of the year is listed below. This includes all of the components purchased as well as testing devices.

<table>
<tr><th>Part</th><th>Material</th><th>Vendor</th><th>Cost</th></tr>
<tr><td>Stage Controller (x3)</td><td>N/A</td><td>Amazon</td><td>$53.39</td></tr>
<tr><td>Stage Top</td><td>PLA</td><td>Alcon</td><td>N/A</td></tr>
<tr><td>Gripper Knobs (x2)</td><td>Nylon 12</td><td>Alcon</td><td>N/A</td></tr>
<tr><td>Motor Holder</td><td>PLA</td><td>Alcon</td><td>N/A</td></tr>
<tr><td>Zeroing Clamp</td><td>PLA</td><td>Alcon</td><td>N/A</td></tr>
<tr><td>Arduino Mega 2560</td><td>N/a</td><td>Amazon</td><td>22.99</td></tr>
<tr><td>Wire 24 AWG(M-F) (x8)</td><td>N/A</td><td>Arduino Starter Kit</td><td rowspan="3">$44.99</td></tr>
<tr><td>Wire AWG (M-M)</td><td>N/A</td><td>Arduino Starter Kit</td></tr>
<tr><td>Breadboard</td><td>N/A</td><td>Arduino Starter Kit</td></tr>
<tr><td>ULN2003 Motor Controller (x2)</td><td>N/A</td><td>Amazon</td><td>$14.99</td></tr>
<tr><td>28BYJ48 Stepper Motor (x2)</td><td>N/A</td><td>Amazon</td><td>$14.79</td></tr>
<tr><td>12V Power Adapter (Barrel Connector)</td><td>N/A</td><td>Amazon</td><td>$10.50</td></tr>
</table>

Barrel Connector to Pos/Neg Adapter	N/A	Amazon	$15.90
Glue	Super glue	Amazon	$7.50
AmScope H800 Microscope	N/A	Microscope Central	$1031.99
AmScope MU-1000 HS Microscope Digital Camera	N/A	Microscope Central	N/A (included with microscope)
Total:	N/A	N/A	$1323.82

Design Timeline

		November				December				January				Febuary				March			
Task #	Task Title	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
1	Discuss Potential Counting Technologies																				
2	Discuss Initial Design Options																				
3	Develop Test / Analysis Methods																				
4	Begin Initial Design Process																				
5	Cost Analysis																				
6	Basic Prototype																				
7	Present Prototype to Mentors																				
8	Revise Prototype I																				

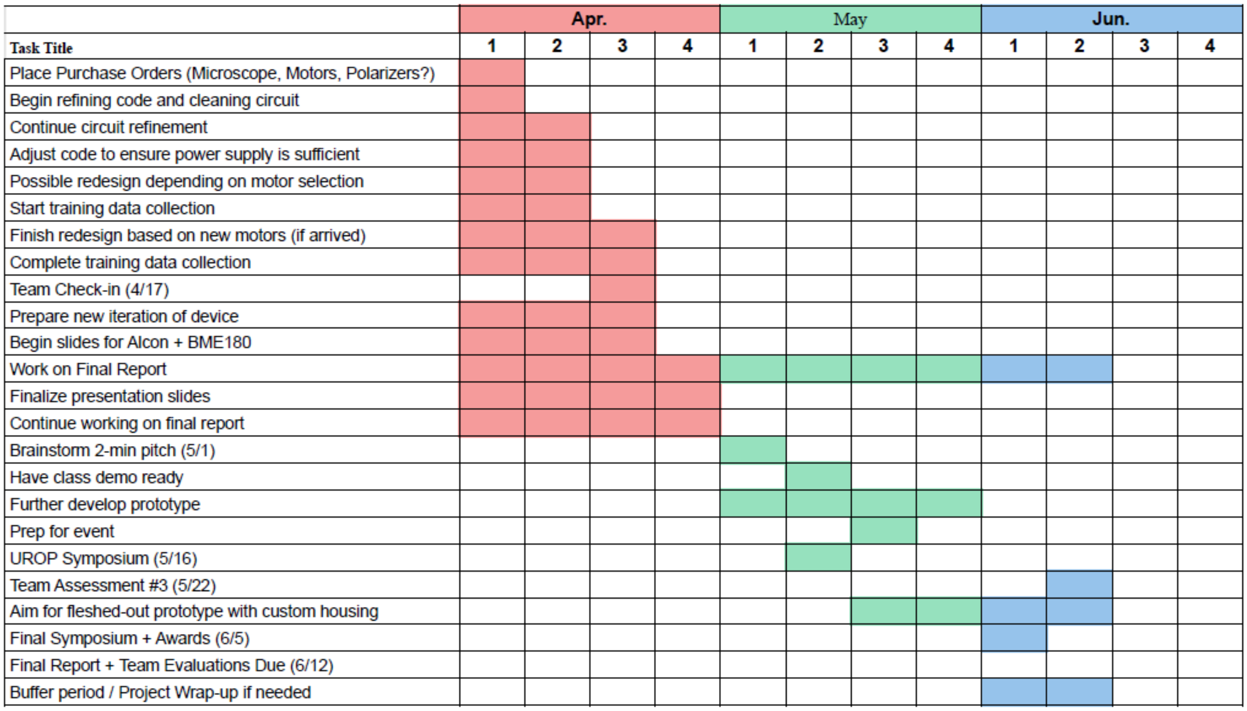


Figure 7. Gantt chart

Manufacturing Documentation

Bill of Materials

Part	Material	Vendor	Cost
Stage Controller	N/A	Amazon	$53.39
Stage Top	PLA	Alcon	N/A
Gripper Knobs (x2)	Nylon 12	Alcon	N/A
Motor Holder	PLA	Alcon	N/A
Zeroing Clamp	PLA	Alcon	N/A

Arduino Mega 2560	N/a	Amazon	22.99
Wire 24 AWG(M-F) (x8)	N/A	Arduino Starter Kit	$44.99
Wire AWG (M-M)	N/A	Arduino Starter Kit	
Breadboard	N/A	Arduino Starter Kit	
ULN2003 Motor Controller (x2)	N/A	Amazon	$14.99
28BYJ48 Stepper Motor (x2)	N/A	Amazon	$14.79
12V Power Adapter (Barrel Connector	N/A	Amazon	$10.50
Barrel Connector to Pos/Neg Adapter	N/A	Amazon	$15.90
Glue	Super glue	Amazon	$7.50
Total:	N/A	N/A	$279.55

Subassemblies

The automated particulate counting system is composed of several integrated subassemblies, each designed for precision, modularity, and ease of use in a production environment. The foundation of the system is a motorized stage assembly, which facilitates controlled, repeatable motion of handpiece filter membranes beneath the imaging unit. This subassembly includes aluminum T-slotted rails, stepper motors with timing belts, and a custom tray designed to securely hold the filter slides. The tray features a side slot, allowing filters to be quickly inserted or removed without disassembling the unit. Stepper motors are connected via knobs to the mount, allowing easy gear swaps and maintenance. The entire motion system is designed to move along a predefined path, ensuring the entire filter surface is captured during scanning.

The camera and microscope subassembly consists of a USB digital microscope mounted above the motorized stage. The mount was aligned manually to center the microscope's field of view on the filter. Fine adjustments are possible through height-tuning screws integrated into the mount. Lighting is provided either through built-in LEDs on the microscope or a fixed external ring light, ensuring consistent illumination across scans. The imaging system plays a critical role in downstream analysis, as consistent, high-contrast images are required for accurate particulate detection.

To coordinate the movement and image capture process, the system incorporates a motion control subassembly consisting of a microcontroller (such as an Arduino) connected to motor drivers and a regulated power supply. Communication between the Python GUI and the hardware is handled via PySerial, which sends G-code-like commands to move the stage. Limit switches and reset routines are implemented in both hardware and software to ensure the system always begins each scan from a known position. This modular control system allows easy adjustment of movement parameters and troubleshooting during development.

At a higher level, all subassemblies are integrated into a complete automated particulate counting system, designed to replace manual inspection processes in Alcon's manufacturing pipeline. The final assembly consists of the motorized scanning platform, imaging unit, and embedded electronics, all mounted on a shared baseplate to ensure structural stability. Cables are routed carefully to minimize interference with moving parts, and the system is housed in an open-frame enclosure for ease of access and prototyping. The hardware components are seamlessly integrated with a Python-based software platform, forming a closed-loop system from physical scan to digital classification.

The technician-facing graphical user interface (GUI), built in Python using PyQt5, enables intuitive control of the entire system. It includes start and stop buttons, live status indicators, and real-time display of the scanned image, particle count, and pass/fail classification. The GUI also maintains a session log with timestamped results and provides an export function for CSV-based data reporting. This software replicates the technician workflow and streamlines quality assurance by automating particulate detection using a convolutional neural network (YOLOv5), applied to each captured image. The results are compared to thresholds derived from ASTM standards, providing consistent and objective assessments.

Throughout the manufacturing process, an emphasis was placed on design for scalability and maintainability. Components such as the filter tray, mounts, and gear mechanisms were prototyped using 3D printing to accelerate iteration cycles. The final design supports tool-less filter exchange, minimizing technician effort. By standardizing hardware mounts and using modular code, the system can be easily upgraded with higher-resolution cameras or alternate detection models. Overall, the system improves throughput, repeatability, and reliability compared to traditional manual inspection, offering a scalable solution for high-volume manufacturing environments.

At the highest level, the device consists of an automatic stage controller, the software program, and the digital microscope. By connecting each 28BYJ-48 stepper motor on the motorized stage subassembly to their respective ULN2003 driver board with male-female square jumper wires, the motorized stage assembly is integrated with the motion control subassembly. The motion control subassembly actuates the stepper motors via these motor controllers, and the stepper motors move the stage accordingly. The motion control subassembly is integrated with the movement component of the software via serial communication commands sent through a USB connection between an Arduino Mega 2560 and a computer.

On this computer, an arduino script that defines the movement commands runs concurrently with the primary python script, enabled by the pyserial library, which sends these movement commands to the stage controller. Simultaneously, the imaging subassembly, consisting of an Amscope H800 microscope and an Amscope MU-1000 HS microscope digital camera, connected to the computer via USB, streams video to the AmLite software.
After every stage movement increment, the Python script screenshots a defined region of the AmLite window and adds the image to a folder. Following completion of the stage controller raster scan, the python script forwards this folder to its YOLOv5 inference module. The resulting particle array, containing centroid coordinates, bounding-box dimensions, and confidence values, closes the loop between motion, imaging, and analysis, thereby integrating all subassemblies into one device.

The prototype can be scaled to high-volume production after several changes to its hardware and software. All 3D-printed components, which include the motor holders, filter tray, and stage-top rails, would be re-engineered for production out of injection-molded plastic or, where higher stiffness and specificity is required, out of machined aluminum. The current breadboard-based Arduino control subassembly would be condensed into a single PCB.

On the imaging side, the H800 microscope and the MU-1000 HS camera would be replaced with the Caltex HD-4K UltraHD microscope and camera system that Alcon currently uses for the microscope particle count test. Images would be extracted directly from the camera software instead of via screenshots. The software itself would be modified into a version-controlled executable software with an installer and calibration routine compatible with Alcon’s ISO 13485 quality-management system.

Critically, the machine-learning pipeline would shift from a YOLOv5 model trained on artificially-generated titanium particles to one trained on an extensive dataset of debris collected from failed production-line slides. This dataset would match the particle morphology that appears in real inspections, thereby equalling technician precision under actual operating conditions.

Because all major components (stepper motors, microscope cameras, power supplies, custom PCBs) are commercially available from multiple suppliers, and because each subassembly can be manufactured in parallel, the system is well suited to high-volume production. With these changes, the design can be manufactured reliably at scale while meeting Alcon’s cost, throughput, and quality requirements.

Materials Validation

The material property requirements of the 3D printed components were minimal. The material needed to be inexpensive, hard enough to not lose its shape over time or be significantly ground down over time, and it needed to provide a good enough tolerance for our components to fit together without any detrimental play.

All of the components are composed of PLA except for the gripper knobs which are made of Nylon 12. PLA was initially chosen for all of the components due to its availability and cost benefits. However, after testing, the gripper knobs showed significant play in its interface with the stage controller. Thus, Nylon 12 was chosen for the material of the gripper knobs for its similarity to PLA, its availability, and its higher printing tolerances that would allow for a tighter interface with the stage controller.

Material testing occurred during the prototype stages. The knobs specifically were tested against the torque of the stepper motors to ensure that they would not shear during normal and heavy operation. The knobs were run through multiple cycles of the device and no shearing or deformation was found. Due to its similar material properties compared to PLA, PLA's strength and resilience to deformation and shear in previous tests, the functional test performed was sufficient to determine the validity of the Nylon 12 used in the gripper knobs.

Data Sheets:
PLA:
https://www.bcn3d.com/wp-content/uploads/2019/09/BCN3D_FILAMENTS_TechnicalDataSheet_PLA_EN.pdf
Nylon 12: https://plastim.co.uk/wp-content/uploads/2019/07/Nylon-12-Technical-Data-Sheet.pdf

The materials were accepted by way of functional tests that determined the strength of the material and its resistance to significant degradation, wear, and deformation. The materials pass if they show no sign of cracks, shear, or significant wear after multiple cycles. The motor holder and stage top do not have any moving parts and are adhered to the stage controller and consequently did not show signs of failing the acceptance criteria. The gripper knobs, in initial testing with PLA, did pass the acceptance criteria but due to the inexact tolerance of the part, Nylon 12 was chosen for future iterations. In testing, Nylon 12 also passed against the same criteria.

The materials passed the acceptance criteria with no signs of imminent failure.

Design Validation

Design Requirements:

Design Requirements	**Justification**
Autonomous	Once the data is collected from our imaging method, we will write a script that would analyze the data to make the particulate analysis autonomous. Pass/Fail
Time-efficient	The current time it takes to analyze each handpiece manually is approximately 18-30 minutes per 10 handpieces.

	A realistic goal for this requirement is approximately 2-3 minutes per 1 handpiece Time < 2-3 minutes per 1 handpiece
Cost-efficient	Prototype needs to be an affordable device for the company to reproduce. Cost < $2000
Identifies particle count per given area and individual size	The basis of the script is to receive imaging data and do the following: ● Quantify the particulate count per given area ● Measure the size of each individual particle Pass/Fail
Determines if handpiece is safe for use, provides feedback, and states the size of largest particle if it is greater than 4.00 mils	The system needs to determine if the handpiece is ready for production given Alcon's testing criteria. To comply with Alcon's particulate testing criteria, we need the system to provide a report on all particulates that were detected with their sizes stated. Pass/Fail
Records necessary information of handpiece if test is passed or records the test number and if handpiece fails	System needs the documentation of each handpiece once it has passed the test. If the test failed, then we need to record the number of testing exposures and determine if the handpiece should be discarded to comply with Alcon's testing criteria. Pass/Fail
Integration of AI	For accurate precision, machine learning software can improve the accuracy of the script we are using to analyze particulate. Add what, how, and importance of AI

	Flowchart of AI components Pass/Fail
Useability	How accessible and easy-to-use is our prototype to technicians. ● Ask technician if there is something in the process that would ease the process, labor wise or time wise ● Survey for technician with prototype - make a measurement where they can show how what we did improved their process Pass/Fail

1: low, 5: high Customer importance rating	Desired direction of improvement (↑,0,↓) Functional Requirements (How's) → Customer Requirements - (What's) ↓	Image input handling and preprocessing	YOLOv5 object detection	Particle measurement	Particle data storage and tracking	Report generation
		↑	↑	↑	↑	↑
5	Particle identification accuracy	1	9			
5	Image processing speed	9	9	9	3	
3	User-friendly interface	9	1	3	3	9
5	Automated particle identification	1	9			
5	Automated particle size measurement	1		9		
3	Generate report	9	3	3	9	9
3	Integration with existing hardware	9	1	1		3
5	Particle size measurement accuracy	1	1	9	3	
5	Store and track particle data (position, size, count)	3	1	3	9	
5	Accurately rejects non-titanium particles	9	9			
	Technical importance score	191	200	156	66	63
	Importance %	25%	26%	21%	9%	8%
	Priorities rank	**2**	**1**	**3**	**4**	**5**

Validation Components

As part of design validation, the Alcon “Validation Assessment” form was completed. Based on the results, which indicated that the process changes would be “high impact” and “medium control” both software and hardware needed to be validated. As a result, both a Computerized System Assessment and a Functional Test document were completed.

Validation Assessment

Initiator Name	Isaac Hoyt
Project Title	Automated Particle Test
Related Engineering Change Plan or Change Control	Particle Count Test
V&V Tracking Number	**962-**V&V #**-**Rev**VA**
Note: For new/change in product component/material, use the Component Risk-Based Assessment and Qualification form, V-QMS-0018735. For product design change, use the Design Control Procedure per V-QMS-0055581.	

<table>
<tr><td colspan="2">Step 1: Description
This step provides relevant information about the system being assessed.</td></tr>
<tr><td rowspan="4">Step 1a
System Description</td><td>System / Change Description:
(1) What is the current situation (provide system description and background)?
(2) What is the proposed change and why is it being proposed (provide the objective and the rationale for the change)?
(3) What is the scope (boundary) of the change? (4) If making a change to an existing system, what is the original validation number?</td></tr>
<tr><td>Current Situation: The current handpiece metal particulate test as explained in the Manufacturing Operating Procedure for Digital Measurement Microscope (V-QMS-0089426) requires technicians to manually count the metal particulates found in the filter membranes leading to a surplus on time invested and potential errors.

Proposed change: To enhance and automate this tedious process, additional software and hardware components will be integrated. The hardware components will aid the software commands to ensure precise functionality. The software will capture and analyze images of the filter membrane to detect metal particulates, with stepper motors controlling movement. Once the measurements are recorded, the images will be deleted, and a report will be generated for pass/fail classification based on a predetermined threshold.

The hardware components consist of 3-D printed mounts to secure the filter membrane and its casing, as well as housings for the stepper motors and knobs designed to fit both the purchased microscope stage and the motors. Additionally, polarizer sheets will be placed over the filter membrane to enhance contrast between the background and the metal particles, improving detection accuracy.

Scope: The change to the metal particulate test will increase production time by speeding up the time it takes to conduct the metal particulate test. By increasing production of the handpiece, the production of the Centurion system would be increased. In addition, it would reduce the need for highly-trained technicians to conduct metal particulate tests, increasing the resources Alcon has.

Validation number: V-QMS-0089427</td></tr>
<tr><td>Intended Use:
(1) Where is the system used (Manufacturing, Receiving Inspection, Complaint Handling, etc.)? (2) What are the associated products / product line(s)? (3) How is the system used to support the product line and how does the change affect usage? (4) What are the associated Quality Attributes? (5) What are the associated process parameters?</td></tr>
<tr><td>Associated Products / Product Line: The Handpiece Metal Particulate Count Test is performed on assembled handpieces within the manufacturing environment, immediately following sensor calibration and pressure testing. This test evaluates the Centurion Ozil, Ozil (Torsional), Infiniti U/S, Active Sentry, and Unity Handpiece, to ensure that assembled handpieces adhere to Alcon’s allowable limits for emitted metal particles.
Use of the System for associated product/product line: Particles leftover from the handpiece shell machining process are caught by the membrane disc filter during the Handpiece Particulate Emission Test, and evaluated to ensure that their number falls within the allowable limit for each Particle Size Range, as seen in Manufacturing Test Procedure for Handpiece Metal Particulate V-QMS-0089427.

Associated (ALL) Quality Attributes: Per 957-1460-001 Rev B, cleanliness per metal particulate test is a critical quality attribute. The 5 minute short cycle is the documented control plan.

Associated (ALL) Process Parameters (as applicable): NA

Provide a brief description of the intended use. Ensure that you are answering the questions above. 😛</td></tr>
</table>

Step 1b System Type	**Impacted System Type:** *(Check all that are directly involved in the change)*			
	☐ Facility / Utility	Step 3.1	☐ Equipment	Step 3.2
	☐ Test Method	Step 3.3	☐ Process [1]	Step 3.4
	☐ Assembly Operations / Line [2]	Step 3.5	☐ Computer System / Software [3]	Step 3.6
	[1] Set of interrelated or interacting activities (e.g., physical, or chemical interactions) that requires specified process parameters (e.g., temperature, heat, pressure, etc.) to transform inputs (raw materials) into outputs (products). For example, Sterilization Process, Packaging Sealing Operations, Injection Molding, Thermoforming Process, Chemical Bonding Process, etc. *[2] An arrangement of machines, tools, and workers in which a product is assembled (e.g., mechanical fasteners (screw, pins, bolts, nuts), snap in place, adhesive joining, etc.) in a particular sequence as it is moved along a defined line or route.* *[3] A computer/software system used within the Manufacture, Testing, DHR, Regulatory Submission or Registration of final or in-process product that is not associated with equipment. A computer system/software that is associated with equipment should use the equipment field.*			

Step 2. GxP (QSR Quality System Regulated**) Assessment**

This step will determine if the system / change impacts a GxP system.

If the result of this assessment is that a GxP system is impacted (new or changed), proceed to the steps(s) identified in Step 1b and complete all steps of this form. Otherwise, proceed to step 2c, identify the deliverables for this system, no further steps are required.

Step 2a	**New or Change to Existing System**			
	New	**Change**		
	☐	☐	Is this a change to an existing GxP (QSR) relevant system or a new system?	**If "New", proceed to step 2b. Otherwise proceed to the step(s) identified in step 1b.**

Step 2b	**New System** *Answer each of the questions below. If any of the questions are yes, this is a GxP (QSR) relevant system and requires further assessment. Proceed to the step(s) identified in step 1b. If all the answers are no, proceed to step 2c.*		
	Yes	**No**	
	☐	☐	Is the system used during the Manufacture, Testing, Labeling, or Packaging of final or in-process product?
	☐	☐	Is this a system used for finished product Cleaning/Sanitization/Sterilization?
	☐	☐	Does the system Create or Maintain environmental conditions to preserve product CQAs (Critical Quality Attribute)?
	☐	☐	Does the system impact CQAs or CPPs (Critical Process Parameter)?
	☐	☐	Does the system Control, Record, and/or update the release status of final or in-process product?
	☐	☐	Does the system store any data to be documented or used in the Device History Record (DHR)?
	☐	☐	Does the system support Regulatory Submission or Registration?
	☐	☐	Is this a computer/software system used within the Manufacture, Testing, DHR, Regulatory Submission or Registration of final or in-process product? *(Software, which is used as a medical device itself, is an accessory to a medical device, or is incorporated as a component in a medical device is out of scope).*
		☐	None of the above (Proceed to Step 2c)

Step 2c	*Complete this step only If the results in steps 2a and 2b indicate that this is not a GxP system; if so, use the table below to designate the deliverables required for this system and submit for review and approval.* *Qualification/validation is NOT required.*	☐ N/A *(This is a GxP (QSR) system that requires further assessment)*

Good Engineering Practice (Select all Applicable)

Note: *This is not an all-inclusive list. Additional deliverables may be applicable*

☐ Design Drawings	☐ Calibration	☐ Preventive Maintenance	☐ Operating Procedures
☐ Safety Assessment	☐ Commissioning	☐ Operator Training	☐ Asset Registration
☐ Spare parts Inventory	☐ Disaster Recovery Plan	☐ Operations & Maintenance Manual	☐ Risk Assessment & Management

☐ Other: Click or tap here to enter text.

Step 3.1: Facility/Utility Assessment *For new or change to Facility/Utilities*	☐ **N/A** *(There are no facility impacts associated with this change)*

Step 3.1a

Qualifications/Validation Risk Assessment Required:

Qualification/Validation risk assessment is required if any of the answers below (step 3.1a) are "Yes". Proceed with the assessment.

Yes	No	
☐	☐	Construction of new facility to support the manufacturing of products where the *facility conditions could affect the product*?
☐	☐	New utility system that contains PLC or software where the *utility conditions could affect the product*?[1]
☐	☐	New controlled environment for the manufacturing of the product?
☐	☐	New facility/utility for sanitation or sterilization process?
☐	☐	Influences/Changes to operating parameters? *(e.g., temperature, humidity, airflow rate, pressure changes)*
☐	☐	Influences/Changes to major components? *(e.g., HVAC, filters, interlocks, pumps, motors, piping, valves, tanks, heat exchangers, disinfection systems)*
☐	☐	Influences/Changes any significant inputs? *(e.g., utility, computer system)*
☐	☐	Influences/Changes in design or configuration? *(e.g., materials of construction, size, layout, distribution systems, components, software configurations)*
☐	☐	Influences/Changes in control systems (PLC or Software)? [1] *(e.g., monitoring instruments, computer systems, user interface, configuration)*
☐	☐	Potential influence to system performance? *(e.g., adding equipment or personnel, production volume increases, product flow, material movement, corrective maintenance, adding/removing use points, increase in demand, system compromised)*

	☐ ☐ Other: Click or tap here to enter text.	
	[1] *Also, Complete Computerized System Validation (CSV) Assessment V-QMS-0019195*	
Comments: Click or tap here to enter text.		☐ N/A

Step 3.2: Equipment Assessment *For new equipment or change to qualified equipment*		☐ **N/A** *(There is no equipment associated with this change)*
Step 3.2a	**Equipment Validation is NOT Required:** *For step 3.2a, if any of these are "Yes", proceed to step 5. Indicate none for the outcome. Provide details in the comment step.*	
	Yes **No**	
	☐ ☐ System requirements are verified through (1) Calibration AND/OR (2) Preventive Maintenance?	
	☐ ☐ Portable or handheld calibrated tool(s) with a single function?	
	☐ ☐ Non-Automated, manual, and simple mechanical or electrical fixture used to aid in manufacturing assembly or inspection or material handling/preparation *(e.g., do not impact product quality attributes)?*	
	☐ ☐ Relocation of the portable/mobile systems (refer to previous validation state/report)?	
Step 3.2b	**Qualifications/Validation Risk Assessment Required:** *For step 3.2b, if any of these are "Yes", a validation risk assessment is required. Proceed with the remainder of this assessment.*	
	Yes **No**	
	☐ ☐ New Equipment for Sterilization cycle or process?	
	☐ ☐ New equipment that generates product labeling?	
	☐ ☐ New sterile packaging equipment?	
	☐ ☐ New equipment that influences controlled environments/areas?	
	☐ ☐ New production equipment with PLC or software? [1]	
	☐ ☐ New test/inspection equipment with PLC or software? [1]	
	☐ ☐ Influences/Changes to operating parameters? *(e.g., Process parameters, installation conditions, speed, flow rate, temperature, pressure)*	
	☐ ☐ Influences/Changes to major components? *(e.g. motors, controllers, software/firmware)*	
	☐ ☐ Influences/Changes to any significant inputs? *(e.g., facility, utility)*	
	☐ ☐ Influences/Changes to any firmware/software/computer systems/user interface? [1]	
	☐ ☐ Influences/Changes in design or configuration? *(e.g., equipment specifications, product contact materials, upgrades, manufacturing materials)*	
	☐ ☐ Influences/Changes in control systems? *(e.g., monitoring instruments)*	
	☐ ☐ Potential impact to system performance? *(e.g. relocation of non-portable equipment, change in environmental conditions, corrective maintenance)*	
	☐ ☐ Replication of previously validated equipment?	

	☐ ☐ Other: Click or tap here to enter text.	
	[1] *Also, Complete Computerized System Validation (CSV) Assessment V-QMS-0019195*	
Comments: Click or tap here to enter text.		☐ N/A

Step 3.3: Test Method Assessment *For new Test method or change to validated Test method*		☐ **N/A** *(There is no test method associated with this change)*
Step 3.3a	**Test Method Validation is NOT required:** *For step 3.3a, if any of these are "Yes", proceed to step 5. Indicate none for the outcome. Provide details in the comment step.*	

Yes	No	
☐	☐	In-process tests for attributes that are subsequently measured with a validated test method (e.g., tests used for process monitoring and control)?
☐	☐	Inspection for attributes that have no patient risk (e.g., cosmetic attributes)?
☐	☐	Visual inspection methods that are self-evident?
☐	☐	Visual inspection methods where operator certification is sufficient to demonstrate repeatability?
☐	☐	Tests that have no parameters or potential sources of variation other than the equipment or instrument utilized, which is controlled through calibration?

Step 3.3b

Qualifications/Validation Risk Assessment Required:
For step 3.3b, if any of these are "Yes", a validation risk assessment is required. Proceed with the remainder of this assessment.

Yes	No	
☐	☐	A new inspection or test process will be used to monitor, inspect, or test critical quality attribute(s) (CQA) of the product?
☐	☐	Test Method used during validation/verification activities?
☐	☐	Test Method used for the in-process and final acceptance production tests?
☐	☐	Test Method used for the incoming inspection testing?
☐	☐	Test Method is based on the compendium (e.g., USP, ASTM, ISO, etc.)?
☐	☐	Influences/Changes to test method validation requirements (e.g., test method parameters, product specifications, the intended use of the method, intended users, intended environment, etc.)?
☐	☐	Influences/Changes to any significant inputs (e.g., facility, utility, test equipment, users, environment, and/or associated computer system)?
☐	☐	Potential impact to test method performance (e.g., changes to work instructions, sample handling/preparation, operators, etc.)?

	☐	☐	Other: Click or tap here to enter text.

Comments: ☐ N/A

The change is to make the metal particulate test automated. Visual inspection is still required for any errors during automated tests. New test will have its own set of instructions for use, but will use current metal particulate testing criteria.

Step 3.4: Process Assessment *For new process or change to validated process*				☐ **N/A** *(There is no process associated with this change)*
Step 3.4a	**Process Validation is NOT Required:** *For step 3.4a, if any of these are "Yes", proceed to step 5. Indicate none for the outcome. Provide details in the comment step*			
	Yes	**No**		
	☐	☐	100% of all process outputs are verified and the established verification step(s) uses a validated test method to confirm if the output is acceptable vs unacceptable?	
Step 3.4b	**Qualifications/Validation Risk Assessment Required:** *For step 3.4b, if any of these are "Yes", a validation risk assessment is required. Proceed with the remainder of this assessment.*			
	Yes	**No**		
	☐	☐	New sterilization cycle or process?	
	☐	☐	New sterile packaging process?	
	☐	☐	Influences/Changes to process requirements? *(e.g., process parameters, quality attributes, product design or configuration, material changes)*	
	☐	☐	Influences/Changes to any significant inputs? *(e.g., facility, utility, equipment, computer system, upstream processes)*	
	☐	☐	Potential impact to process performance? *(e.g., changes to work instructions, personnel, or process flow)*	
	☐	☐	Other: Click or tap here to enter text.	

Comments: ☐ N/A

Click or tap here to enter text.

Step 3.5: Assembly Operations/Line *For new or change to existing Assembly Line or Re-layout of Assembly Line*		☐ **N/A** *(There is no assembly operation or line associated with this change)*
Step 3.5a	**Qualifications/Validation Risk Assessment Required:** *For step 3.5a, if any of these are "Yes", a validation risk assessment is required. Proceed with the remainder of this assessment*	

Yes	No	
☐	☐	Introduction/ transfer of product into ITC Manufacturing?
☐	☐	Relocation of a Manufacturing Line?
☐	☐	Influences/Changes to Assembly operations/line? *(e.g., area re-layout, sequence change, method change, new/upgraded equipment introduction, new component introduction etc.)*
☐	☐	Influences/Changes to any significant product inputs? *(e.g., product software upgrade, product component change, equipment, etc.)*
☐	☐	Influences/Changes to Assembly Operations Procedure? *(e.g., MAP, MOP, etc.)*
☐	☐	Changes potentially impacting the performance of the Assembly line? *(e.g., changes to work instructions, personnel, or process flow)*
☐	☐	Other: Click or tap here to enter text.

Comments: ☐ N/A
Click or tap here to enter text.

Step 3.6: Computerized System / Software
For new software implementations or changes to existing software systems

☐ **N/A**
(A computerized system is not affected with this change)

Note: *GxP relevant computerized systems are* ***required*** *to be added to the site Master Validation Plan.*

Step 3.6a — **System Type, Complexity and Risk Assessment**

For step3.6a, if "Yes" provide the Classifeye ID and proceed to Step 3.6.b.

Note: *The Impacted system type and complexity may be determined through a Classifeye assessment (recommended) or V-QMS-0019195 (Computerized System Validation assessment).*

Yes	No		
☐	☐	Is or has a Classifeye assessment been performed for this system? **Classifeye ID#:** Click or tap here to enter text.	**If yes, provide the Classifeye ID.** / **If no, V-QMS-0019195 is required.**

Step 3.6b — **Usage of Computer Software Assurance Program**

For step3.6b, if "Yes", the risk assessment will be performed in the URRA and further assessment in this form is not required. Document the URRA number, location (e.g. box, Windchill, etc.), provide a summary of the URRA outcome, and proceed to step 6.

Yes	No		
☐	☐	Will the qualification activities utilize the Computer Software Assurance (CSA) program? **URRA ID#:** Click or tap here to enter text. **URRA Location:** Click or tap here to enter text.	**If yes, provide the URRA ID, location, and summary. Proceed to step 6.** / **If no, V-QMS-0019195 is required proceed to step 3.6c.**

URRA Summary
Consider the following:
(1) Is this new code or a change to existing? (2) What features / requirements are impacted?
(3) What is the resulting test level? (4) What is the impact to the existing workflows and associated product?

	Click or tap here to enter text. ☐ N/A	
Step 3.6c	**Qualifications/Validation Risk Assessment Required:** *For step 3.6c, if any of these are "Yes", a validation risk assessment is required. Proceed with the remainder of this assessment*	
	Yes **No** ☐ ☐ New software system that directly supports a process or manages data that has impact to product quality and patient safety? ☐ ☐ Influences/Changes any significant inputs? ☐ ☐ Influences/Changes in design or configuration? ☐ ☐ Influences/Changes in control systems? ☐ ☐ Potential influence to system performance? ☐ ☐ Other: Click or tap here to enter text.	
Comments: ☐ N/A Click or tap here to enter text.		

Step 4	**Risk Assessment / Classification**			☐ **N/A** *(Risk Assessment is not required)*
Step 4a:	**Impact Assessment**: Identify the potential impact associated with this system: Consider answering the following questions: ***(1)*** *What are the associated Quality Attributes?* ***(2)*** *What are the associated process parameters?* ***(3)*** *Is there any impact to the Patient / User?* ***(4)*** *Is there an impact to Regulatory/Compliance?* ***(5)*** *Impact to the functionality of the product?* ***(6)*** *Impact to the business (e.g. Disruption in manufacturing operations)?*			
	Review & Reference V-QMS-0019513 and / or RTM/PRTM, RMR, FMEA, System Specifications as a supporting document (as applicable) Quality Attributes: Affordability, autonomy, convenience, efficiency, precision, and usability Process Parameters: Time, particle size Impact to User: Easier useability, less labor intensive Impact to Patient: If product passes with false-positive, patient health at risk Impact to Compliance: Potential oversights will make product unsafe for use, direct impact on patient safety Impact to Functionality of Product: Product may be unsafe for use if false-positive occurs Impact to Business: Product recall if product passes with false-positive, disruption in quality assurance process			
	Result	☐ Low Impact	☐ Medium Impact	☐ High Impact
	Example	*No impact on patient, user, or their safety. No impact on final product functionality. The impact is non-functional or cosmetic only. The impact would not result in significant production cost or field correction.*	*Direct impact on functionality of the product/system. The system may generate products that result in customer dissatisfaction or significant production cost (high rejects/ yield) or field corrections. Direct impact on regulatory compliance. There are no potential hazards to patient/user safety.*	*Direct impact on product quality, patient/user, or patient safety. May result in a field action alert or product recall.*

<table>
<tr><td rowspan="4">Step 4b:</td><td colspan="4">Control Level: Assess the effectiveness of existing controls

(e.g., Inspection and/or documentation from supplier, Receiving Inspection (RI), In-Process testing or verifications, Downstream Testing, Final Testing, Manufacturing Test Procedure (MTP), Manufacturing Operating Procedure (MOP), Manufacturing Assembly Procedure (MAP), and Documented Study/analysis, etc.)</td></tr>
<tr><td colspan="4">Rationale (Include discussion of inspection method and controlled document references with document number, revision, and step number):
There is no downstream check needed for this test. However, we need one test or inspection for the handpiece.</td></tr>
<tr><td>Result</td><td>☐ Low Control</td><td>☐ Medium Control</td><td>☐ High Control</td></tr>
<tr><td>Example</td><td>The subsequent tests or inspection are not effective, or nonexistent to inspect, or test for all the identified impact and hazards.</td><td>All relevant requirements are partially verified by established in-process testing, verification, or downstream test. The effectiveness of the downstream test has the potential for latent or intermittent defects (misses). Example: (1) Inspection using sampling (not 100%) of the product. (2) Inspection using a non-validated test method</td><td>The identified inspection step is 100% effective to inspect or test for all the identified impacts and hazards. e.g.:
1. 100% downstream check: 100% of all units are inspected/tested, where appropriate, using validated test method.
2. 100% in-process testing or verification.</td></tr>
</table>

<table>
<tr><td rowspan="6">Step 5</td><td colspan="4">Recommended Strategy: Final Strategy is documented in Step 6.</td></tr>
<tr><td>Control Level →
Impact ↓</td><td>Low Control</td><td>Medium Control</td><td>High Control</td></tr>
<tr><td>Low Impact</td><td>☐ Engineering Study [1]
(Unscripted Testing)</td><td>☐ Engineering Study [1]
(Unscripted Testing)</td><td>☐ None</td></tr>
<tr><td>Medium Impact</td><td>☐ Functional Test [1]
(Scripted Testing)</td><td>☐ Functional Test [1]
(Scripted Testing)</td><td>☐ Engineering Study [1]
(Unscripted Testing)</td></tr>
<tr><td>High Impact</td><td>☐ Validation/Qualification[2]
(Scripted Robust)</td><td>☐ Validation/Qualification[2]
(Scripted Robust)</td><td>☐ Functional Test [1]
(Scripted Testing)</td></tr>
<tr><td colspan="4">1 = Provide details for the purpose, scope, and objective/goal in the comment step below.
2 = Review revalidation/requalification requirements per V-QMS-0018991, and update Site Validation Master Plan as required.</td></tr>
<tr><td colspan="5">Comments: ☐ N/A
Because this qualification test determines if the handpiece is safe for use, patients' health is at risk. This is a high impact test</td></tr>
</table>

Step 6	**Deliverables (refer to V-QMS-0018245)** *Identify the appropriate deliverables for the qualification activities*		
Validation Life-Cycle Phase	**Required Elements**	**Required?**	**Comments**
Planning	Change Control / Change Plan	☐	Click here to enter text.
	Validation Plan	☐	Click here to enter text.
Pre-Qualification	Fixture Specifications (FS)	☐	Fixtures to be added include stage controller and our 3d printed controls.
	User Requirement Specifications (URS)	☐	Click here to enter text.
	Functional Requirement Specifications (FRS)	☐	Click here to enter text.
	Detailed Design Specifications (DDS)	☐	Click here to enter text.
	User Requirements Risk Assessment (URRA)	☐	Click here to enter text.
	Add to Calibration Management System & Procedure	☐	Click here to enter text.
	Add to Maintenance Management System & PM Procedure	☐	Click here to enter text.
	Test Accuracy Ratio (TAR) for Measurement System	☐	Click here to enter text.
	Manufacturing Software Specifications (MSS)	☐	Click here to enter text.
	Machine Operating Procedures (MOP)	☐	Click here to enter text.
	Manufacturing Test Procedures (MTP)	☐	Automated Particulate Counting Procedure will require new test procedure.
	Manufacturing Assembly Procedures (MAP)	☐	Click here to enter text.
	Requirement Traceability Matrix (RTM) (e.g., PRTM)	☐	Click here to enter text.
	HSE Risk Assessment and Management Plan (V-QMS-0019504)	☐	Click here to enter text.
	Safety Design Guidelines for Custom Manufacturing and Test Equipment (V-QMS-0017589)	☐	Click here to enter text.
	Supplier Approval	☐	Click here to enter text.
	ER/ESIG/DI Assessment (V-QMS-0019195)	☐	Click here to enter text.
	Risk Assessment (FMEA)	☐	Click here to enter text.
	Software Code Review	☐	Software used in Automated Counting Process will need to be reviewed and validated in addition to hardware components.
	Design Review	☐	Overall design will be reviewed before implementation.
Qualification	Facility / Utility ☐ IQ ☐ OQ ☐ PQ	☐	Click here to enter text.
	Equipment / System ☐ IQ ☐ OQ ☐ PQ	☐	Click here to enter text.
	Process ☐ OQ ☐ PQ	☐	Click here to enter text.
	Assembly Operation/Line Verification (AOV))	☐	Click here to enter text.
	Product PQ (PPQ)	☐	Click here to enter text.
	Test Method Validation (TMV)	☐	Current test method validations are sufficient but will need to be mentioned before implementation. i.e. How many particles are generally found on filter, risk of particles to patient health.
	Functional Test	☐	Functional Test to determine that Automated Particulate Test meets existing standards. Automated results will be tested against technician results.
	Engineering study	☐	Click here to enter text.
	Computer Software Assurance (CSA) *specify environments* ☐ IQ ☐ OQ ☐ PQ	☐	Click here to enter text.
Project Closure	Validation Summary	☐	Click here to enter text.
	Update to the Site Validation Master Plan (sVMP)	☐	Click here to enter text.
Retirement	Retirement Plan	☐	Click here to enter text.
	Retirement Summary	☐	Click here to enter text.

APPROVALS ELECTRONICALLY THROUGH WINDCHILL (Minimum required signatures are identified in Validation Approval Matrix located on MS&T SharePoint Drive or Shared Drive)

Computerized System Assessment

Initiator Name	Click or tap here to enter text.
Project Title	Click or tap here to enter text.
Related Engineering Change Plan or Change Control	Click or tap here to enter text.
V&V Tracking Number	**962-**V&V #**-**Rev**VA**

<table>
<tr><td colspan="2">Step 1: Description
This step provides relevant information about the system being assessed.</td></tr>
<tr><td rowspan="3">Step 1a
System Description</td><td>System / Change Description:
(1) What is the current situation (provide system description and background)?
(2) What is the proposed change and why is it being proposed (provide the objective and the rationale for the change)?
(3) What is the scope (boundary) of the change?
(4) If making a change to an existing system, what is the original validation number?</td></tr>
<tr><td>1) Alcon's current particulate counting process is manual, tedious and time consuming, there is not an integrated software that counts and measures the particles. There is only software that processes information about the particles given technician's input and provides a pass/fail output.

2) The team proposes a software for object detection to automatically identify the titanium particles called "YOLO" which stands for "you only look once," this will help automate the counting process by having the software identify the particles for technicians automatically.

3) Scope: The change to the metal particulate test will increase production time by speeding up the time it takes to conduct the metal particulate test. By increasing production of the handpiece, the production of the Centurion system would be increased. In addition, it would reduce the need for highly-trained technicians to conduct metal particulate tests, increasing the resources Alcon has.

4) The scope of the change will be restricted to the methods found in V-QMS-0089427, which describes the operation of Alcon's microscope for inspecting Metal Particle Test filter slides. The change will not modify the quality attributes, acceptance criteria, or prior steps of the Metal Particulate Test.
(correct me if wrong)</td></tr>
<tr><td>Intended Use:
(1) Where is the system used (Manufacturing, Receiving Inspection, Complaint Handling, etc.)?
(2) What are the associated products / product line(s)?
(3) How is the system used to support the product line and how does the change affect usage?
(4) What are the associated Quality Attributes?
(5) What are the associated process parameters?</td></tr>
</table>

1) The system will be used in the manufacturing line
2) Associated Products / Product Line: The Handpiece Metal Particulate Count Test is performed on assembled handpieces within the manufacturing environment, immediately following sensor calibration and pressure testing. This test evaluates the Centurion Ozil, Ozil (Torsional), Infiniti U/S, Active Sentry, and Unity Handpiece, to ensure that assembled handpieces adhere to Alcon's allowable limits for emitted metal particles.
3) Use of the System for associated product/product line: Particles leftover from the handpiece shell machining process are caught by the membrane disc filter during the Handpiece Particulate Emission Test, and evaluated to ensure that their number falls within the allowable limit for each Particle Size Range, as seen in Manufacturing Test Procedure for Handpiece Metal Particulate V-QMS-0089427.
4) Associated (ALL) Quality Attributes: There are two critical quality attributes (CQAs) associated with the metal particulate testing process: CQA ID 3.8 Cleanliness (Metal Particulate) and CQA ID 3.10 Fluid Path Clogged (HP Test). However, the implementation of this change does not impact relevant CQAs
5) Associated (ALL) Process Parameters (as applicable): There are two process parameters associated with the metal particulate testing process: the "Metal Particulate Test" and "Flow rate". However, the implementation of this change does not impact relevant process parameters.

Step 1b **System Type**	**Where Used:** *(Check all that are directly involved in the change)*
	☐ Manufacturing — Product Line: *Handpiece*
	☐ Receiving Inspection
	☐ Complaint Handling
	☐ Other — *Click or tap here to enter text*
Step 1c **System Category**	**System Category:** *Utilize the table below to determine the type of system being implemented*

		Description	**Examples**
☐	**Category 1** Infrastructure	Established or commercial layered software i.e., upon which applications are built. Infrastructure software, system and tools supporting CS life cycle activities and IT and infrastructure processes.	Operating systems – MS windows, Linux, etc. Database manager – MySQL, Oracle, etc. Programming Language – C++, LabVIEW, etc. Logic Ladder Interpreters – Gx Developer, etc. Statistical Tools – Minitab, Stata, etc.
☐	**Category 3** COTS Non configurable software	Non-configuration or limited configuration using factory provided default values or ranges i.e., Parameterization Runtime parameters may be entered and stored but the software cannot be configured to suit the business processes	Firmware based app – Printer, Scanner etc. COTS software – Autocon, MATLAB, etc. Lab Instruments – pH meter, Analytical balance, etc.
☐	**Category 4** COTS Configurable software	Configuration (no code): Making changes to the software's default settings to meet specific business requirements without modifying the source code e.g. • Changing language, time zone & currency • Setting standard security parameters setting up 'user's role and preferences	COTS Production Equipment – SMT, Laser Markers, Batch Cleaners, etc. SCADA – FactoryTalk view, Indusoft, etc. Spreadsheets – MS Excel, Google Sheets, etc. ERP/MES Software – SAP, Opcenter, etc. IEC61131 – Functional Block Diagram application
☐	**Category 5** Custom software	Customization (only code): Writing new code (scripts, programs, class files) within the software to meet specific business requirements e.g. • Alcon developed: o Enhancement in existing features/functions o Additional Features / Functions Add / Delete steps in process	C++, C#, LabVIEW application – ATE's SCADA scripting – Supervisory Control and Data Acquisition IEC61131 – Ladder diagram, sequential function chart, Spreadsheet application – macros

Step 2. Data Integrity (DI) / Electronic Record, Electronic Signature (ERES) Assessment *This step will determine if the system has DI / ERES relevancy.*			
Step 2a	**System Change** *For step 2a, if any of the following questions are "Yes", Data Integrity, Electronic Record and Electronic Signature impact assessment is required. Proceed with steps 3 and 4.* *If all questions are "No" proceed to step 5.*		
	Yes	**No**	
	☐	☐	Does the system create, maintain, archive, retrieve or transmit e-records/electronic signatures?
	☐	☐	Does the system collect Electronic Data (e.g., results, data entered by operator (e.g., parameters))?
	☐	☐	Does the system collect Transient Data (e.g., Data that is transmitted to another computerized system)?
	☐	☐	Does the system create data but not saved electronically (e.g., data is printed or transcribed on a form)?
	☐	☐	Other: *Click or tap here to enter text*

Step 3: Data Integrity Assessment					☐ **N/A** *(DI is not impacted)*
Step 3a	**Yes**	**No**	**Requirement**	**Detailed Description** ***(1) How system meets the requirement*** ***OR*** ***(2) Why the requirement is not applicable***	**Citation**
	☐	☐	System must require data to be Attributable to the person generating the data.	*Click or tap here to enter text*	MHRA DI Definitions & Guidance v1.1: ALCOA
	☐	☐	System must display or print data in a Legible (human-readable) format throughout its retention period.	*Click or tap here to enter text*	MHRA DI Definitions & Guidance v1.1: ALCOA
	☐	☐	System must be designed to ensure that the execution of critical operations are recorded Contemporaneously (at the time of execution) by the user.	*Click or tap here to enter text*	MHRA DI Definitions & Guidance v1.1: ALCOA
	☐	☐	System must ensure that Original data remains accessible and readable throughout the retention period of the data.	*Click or tap here to enter text*	MHRA DI Definitions & Guidance v1.1: ALCOA
	☐	☐	System must Accurately save data at the time the data is generated and in a manner that prevents the original data from be altered, obscured, or deleted.	*Click or tap here to enter text*	MHRA DI Definitions & Guidance v1.1: ALCOA
	☐	☐	Data must be retained in a dynamic form where this is critical to its integrity or later verification.	*Click or tap here to enter text*	MHRA DI Definitions & Guidance v1.1: Primary Record

	Yes	No	Requirement	Detailed Description	Citation
	☐	☐	System must maintain the original data when technically feasible to do so. Otherwise, data must be output as a report validated as a true and accurate representation of all original data and metadata, one that preserves the integrity (accuracy, completeness, content and meaning) of the record.	*Click or tap here to enter text*	MHRA DI Definitions & Guidance v1.1: Original Record / True Copy
	☐	☐	System must assure long term, permanent retention of completed data and relevant metadata in its final form for the purposes of reconstruction of the process or activity.	*Click or tap here to enter text*	MHRA DI Definitions & Guidance v1.1: Data Retention - Archival
	☐	☐	System must assure that a copy of current data, metadata and system configuration settings be maintained for the purpose of disaster recovery.	*Click or tap here to enter text*	MHRA DI Definitions & Guidance v1.1: Data Retention - Backup
	☐	☐	System/Data is stored at the centralized location that can be maintain by the appropriate authorized function.	*Click or tap here to enter text*	MHRA DI Definitions & Guidance v1.1: Data Retention - Backup

Comments: ☐ N/A

Click or tap here to enter text.

Step 4: Electronic Record / Electronic Signature (21 CFR Part 11) Assessment					☐ **N/A** *(ERES is not impacted)*
Step 4a	**Yes**	**No**	**Requirement**	**Detailed Description** ***(1) How system meets the requirement OR (2) Why the requirement is not applicable***	**Citation**
	☐	☐	The system will be able to detect invalid records where applicable (e.g., invalid field entries, fields left blank that should contain data, values outside of limits).	*Click or tap here to enter text*	21 CFR Part 11.10(a)
	☐	☐	The system will allow viewing of the entire contents of the records.	*Click or tap here to enter text*	21 CFR Part 11.10(b)
	☐	☐	The system allows printing of the entire contents of the records.	*Click or tap here to enter text*	21 CFR Part 11.10(b)

	Yes	No	Requirement	Detailed Description (1) How system meets the requirement OR (2) Why the requirement is not applicable	Citation
	☐	☐	The systems allow generation of all records electronically in a format that can be put on a portable medium (e.g., diskette or CD) or transferred electronically	Click or tap here to enter text	21 CFR Part 11.10(b)
	☐	☐	The system will ensure that only authorized individuals can use the system.	Click or tap here to enter text	21 CFR Part 11.10(d)
	☐	☐	The system will allow for different levels of access based on user responsibilities (e.g., user, administrator).	Click or tap here to enter text	21 CFR Part 11.10(l)
	☐	☐	If the system requires sequenced steps, the system must include operational checks to ensure that the actions are performed in the correct sequence.	Click or tap here to enter text	21 CFR Part 11.10(f)
	☐	☐	If it is a requirement of the system that data input or instructions can only come from specific input devices (e.g., instruments, terminals); the system must check for the correct device.	Click or tap here to enter text	21 CFR Part 11.10(h)
	Yes	**No**	**Requirement**	**Detailed Description** *(1) How system meets the requirement* *OR* *(2) Why the requirement is not applicable*	**Citation**
	☐	☐	The electronic records must be protected against intentional or accidental modification or deletion.	Click or tap here to enter text	21 CFR Part 11.10(c)
	☐	☐	If the data is archived off the system, the audit trail must be archived as well and all of the archived data must be accurately retrieved even after system upgrades.	Click or tap here to enter text	21 CFR Part 11.10(e)
	☐	☐	An electronic audit trail function must be automatically generated for all operator entries.	Click or tap here to enter text	21 CFR Part 11.10(e)
	☐	☐	The audit trail must completely be outside the control and access of users (except for read-only access of the audit trail file).	Click or tap here to enter text	21 CFR Part 11.10(e)
	☐	☐	It must be impossible for the user to disable the audit trail function.	Click or tap here to enter text	21 CFR Part 11.10(e)

	☐	☐	The operatorname, date, time, and indication of record (or file) creation, modification must be recorded in the audit trail.	*Click or tap here to enter text*	21 CFR Part 11.10(e)
	☐	☐	If the predicate regulation requires it, the reason for a change must be included in the audit trail.	*Click or tap here to enter text*	21 CFR Part 11.10(e)
	☐	☐	The system date and time must be protected from unauthorized change.	*Click or tap here to enter text*	21 CFR Part 11.10(e)
	☐	☐	When data is changed or deleted, all of the previous values must be electronically available.	*Click or tap here to enter text*	21 CFR Part 11.10(e)
	☐	☐	The audit trail data must be protected from accidental or intentional modification or deletion (read-only access).	*Click or tap here to enter text*	21 CFR Part 11.10(e)
	☐	☐	The electronic audit trails must be maintained and retrievable for at least as long as its respective electronic records.	*Click or tap here to enter text*	21 CFR Part 11.10(e)
	☐	☐	The electronic audit trails must be readily available for inspections and audits.	*Click or tap here to enter text*	21 CFR Part 11.10(e)
	Yes	**No**	**Requirement**	**Detailed Description** ***(1) How system meets the requirement*** ***OR*** ***(2) Why the requirement is not applicable***	**Citation**
	☐	☐	Selected portions of the audit trail must have the capability to be viewed and printed by inspectors.	*Click or tap here to enter text*	21 CFR Part 11.10(e)
	☐	☐	Selected portions of the audit trail must be capable of being extracted in a transportable electronic format that can be read by regulatory agencies.	*Click or tap here to enter text*	21 CFR Part 11.10(e)
	☐	☐	There must be documentation to show that persons who develop the system have the education, training, and experience to perform their assigned tasks (Including temporary and contract staff).	*Click or tap here to enter text*	21 CFR Part 11.10(i)
	☐	☐	There must be documentation to show that persons who maintain the system have the education, training, and experience to perform their assigned tasks (including temporary and contract staff).	*Click or tap here to enter text*	21 CFR Part 11.10(i)

	☐	☐	There must be documentation to show that persons who use the system have the education, training, and experience to perform their assigned tasks (including temporary and contract staff)	*Click or tap here to enter text*	21 CFR Part 11.10(i)
	☐	☐	The distribution of, access to, and use of systems operation and maintenance documentation must be controlled.	*Click or tap here to enter text*	21 CFR Part 11.10(k1)
	☐	☐	Access to "sensitive" systems documentation must be restricted e.g., network security documentation, system access documentation.	*Click or tap here to enter text*	21 CFR Part 11.10(k1)
	☐	☐	"Open" system must employ controls designed to ensure the authenticity, integrity, and, as appropriate, the confidentiality of electronic records from the point of their creation to the point of their receipt and include additional measures such as document encryption and use of appropriate digital signature standards.	*Click or tap here to enter text*	21 CFR Part 11.30
	☐	☐	All electronically signed records must contain the following information associated with the signing: 1) full printed name of the signer (2) date and time of signing (3) meaning of signature (e.g., review, approval).	*Click or tap here to enter text*	21 CFR Part 11.50(a)
	☐	☐	The date and time stamps must be applied automatically (vs. keyed in by the user).	*Click or tap here to enter text*	21 CFR Part 11.50(b)
	☐	☐	Date and time stamps must be derived in a consistent way in order to be able to reconstruct the sequence of events.	*Click or tap here to enter text*	21 CFR Part 11.50(b)
Comments:					☐ N/A

Step 5	**Risk Assessment / Classification**
Step 5a:	**Impact Assessment** *Identify the potential impact associated with this system:* Consider answering the following questions: ***(1)*** *What are the associated Quality Attributes?* ***(2)*** *What are the associated process parameters?* ***(3)*** *Is there any impact to the Patient / User?* ***(4)*** *Is there an impact to Regulatory/Compliance?* ***(5)*** *Impact to the functionality of the product?* ***(6)*** *Impact to the business (e.g., Disruption in manufacturing operations)?*
	Review & Reference V-QMS-0019513 and / or RTM/PRTM, RMR, FMEA, System Specifications as a supporting document (as applicable)

	1) Quality Attributes: There are two critical quality attributes (CQAs) associated with the metal particulate testing process: CQA ID 3.8 Cleanliness (Metal Particulate) and CQA ID 3.10 Fluid Path Clogged (HP Test). However, the implementation of this change does not impact relevant CQAs. 2) Process Parameters: Technician touch time, time to finish work order of ten filter slides, microscope zoom, pixel to mil conversion, microscope focus, detection threshold size (Only objects ≥ 0.50 mil are counted), size-bin limits (0.50–<1.00 mil ≤ 50, 1.00–<2.00 mil ≤ 5, 2.00–<3.00 mil ≤ 3, 3.00–<4.00 mil ≤ 1, ≥ 4.00 mil = 0), sample plan fraction (20% or 100%) 3) Impact to User: Easier useability, less labor intensive, allows users to set up and supervise multiple concurrent particle count tests. Impact to Patient: If product passes with false-negative (fails to identify titanium particles), patient health at risk 4) Impact to Compliance: Potential oversights will make product unsafe for use, direct impact on patient safety 5) Impact to Functionality of Product: Product may be unsafe for use if false-positive occurs 6) Impact to Business: Product recall if product passes with false-positive, disruption in quality assurance process
Step 5b:	**Control Level** *Assess the effectiveness of existing controls:* *(e.g., Inspection and/or documentation from supplier, Receiving Inspection (RI), In-Process testing or verifications, Downstream Testing, Final Testing, Manufacturing Test Procedure (MTP), Manufacturing Operating Procedure (MOP), Manufacturing Assembly Procedure (MAP), and Documented Study/analysis, etc.)*
	Rationale (Include discussion of inspection method and controlled document references with document number, revision, and step number): The software is designed to generate a pass/fail outcome based on detected particle sizes, without currently employing a downstream validation process for software responses. This absence of downstream validation limits the team's control over the software outputs.

Comments: ☐ N/A
Click or tap here to enter text.

Step 6	**Deliverables (refer to V-QMS-0018245)** *Identify the appropriate deliverables for the qualification activities*		
	Required Elements	**Required?**	**Comments**
	Supplier Assessment Verification	☐	N/A
	Hardware Installation Verification *(e.g. Computer System, PLC Hardware, etc.)*	☐	Ensures cameras, computers, the hardware components are correctly installed and ready to function.
	Software/Application Installation Verification *(e.g. PLC Firmware, Computer Software/Application, etc.)*	☐	Validates that the correct software and versions are installed.
	Network Connection Verification	☐	N/A
	Software Configuration Baseline Verification	☐	TBD (repeatability purposes)
	Security Access Verification	☐	N/A
	Object Freeze Verification	☐	N/A
	System Functional Test Verification	☐	verifies the full system performs as expected./
	Backup and Restore Verification	☐	N/A
	System Print Report Verification	☐	N/A

Step 6	Deliverables (refer to V-QMS-0018245) *Identify the appropriate deliverables for the qualification activities*		
	Required Elements	**Required?**	**Comments**
	Alarms, Safety device and Interlock Mode Verification	☐	N/A
	Operator Interface/Command Functions Verification	☐	Users will be operating the software
	Input / Output Verification	☐	Ensures inputs and outputs work correctly.
	Critical Algorithms and Calculation Verification	☐	required to verify that measurements and pass/fail logic are correct.
	Decision Point Verification	☐	TBD
	Camera Decision Function Verification	☐	Ensures cameras detect and process images correctly
	Archiving Verification	☐	N/A
	Audit Trail Verification	☐	N/A
	Other 1: Click here to enter text.	☐	Click here to enter text.
	Other 2: Click here to enter text.	☐	Click here to enter text.
	Other 3: Click here to enter text.	☐	Click here to enter text.
	Other 4: Click here to enter text.	☐	Click here to enter text.
	Other 5: Click here to enter text.	☐	Click here to enter text.
Comments: Click or tap here to enter text.			☐ N/A

APPROVALS ELECTRONICALLY THROUGH WINDCHILL (Minimum required signatures are identified in Validation Approval Matrix located on MS&T SharePoint Drive or Shared Drive)

Functional Test

Test Cases:

Test No.	Procedure	Acceptance Criteria	Result	Pass / Fail	Initial / Date
	Requirement type: How is the requirement to be verified? **Start-up & shut-down:** Prove the system powers up, initialises and shuts down without errors. Attempt same start-up procedure five times.	Requirement specification/acceptance criteria System powers up with no errors. Successful start-up every time, with limited adjustments for stage controller location, serial communication establishment.	5/5	Pass	IH

	Parameter setting: Confirm size-bin limits, pixel-per-mil constant, serial connection port, model path, movement parameters. Test each parameter by changing values and running the code.	All five major parameters can be edited without breaking the program.	5/5	Pass	IH
3.	**Data capture & calculation:** Show one end-to-end run generates correct counts, pass/fail, and report. Run the process and check for normal function.	Functions and outputs results normally.	Pass	Pass	IH
4.	**Error handling** Verify system detects and handles top high-risk failure modes from FMEA.	Most failure modes not integrated into software. Software detects failed model path and masks blue area at margins of filter, but fails to detect most other potential failure modes.	2/5	Fail	IH
5.	**Maintenance hooks (cleaning, calibration)** Demonstrate calibration & cleaning reminders fire at correct interval.	Not added to software.	0	Fail	IH
6.	**Cycle time Comparison** Compares device average cycle time to technician average cycle time.	Technician cycle time varies, but on average about four minutes per filter. Automatic counter cycle time is currently 2 minutes 15 seconds, but could change with further development.	Faster than technician		
7.	**Accuracy Comparison** Verify device counts and measurements match technician gold standard within tolerance.	Instead of comparing with technician, comparison was made between manual counting and YOLO identification for five filter slide images. Device consistently detects and accurately measures particles above 2 mils, but small particles often go undetected. Because YOLO model wasn't trained on genuine manufacturing debris, actual implementation would	~95% accuracy	Fail	IH

		likely be more accurate, especially with smaller FOV for analyzed images.			
8.	**Concurrent device oversight** Confirm one operator can supervise multiple devices without accuracy loss.	N/A	N/A	N/A	N/A

Three "comparison tests" would be most important for determining the suitability of the AIcount device for use in actual manufacturing. These tests would compare the device to the technician in terms of speed, accuracy, and potential for simultaneous testing.

The AIcount device has the same cycle time every time it conducts the particle count test. The current time is 2 minutes and 15 seconds. To compare this time to the average technician inspection time, a time study was conducted. Technician inspection times vary significantly, but on average they took about four minutes. The AIcount device is more time efficient.

To test the device's accuracy, a comparison was made between the YOLO model's performance and manual particle counts. The YOLO model consistently and accurately detects and measures most particles, but it can miss particles measuring 0.5-2 mils. This performance deficit likely stems from the size of the section of filter captured in each image. If the system was trained and inferred on smaller sections of the filter, small particle detection would likely be much better. Concurrent device oversight could not be tested because we only had one device.

The current design does not meet every design criteria. The device is autonomous, time-efficient, and cost efficient, but it does not currently match technician accuracy, especially when identifying very small particles. The device also does not match the usability design criteria. The device is currently too finicky for a technician to consistently set it up and run it with less effort than the current testing method.

In all five accuracy tests, the device failed to detect many of the smallest titanium particles. This fails the accuracy requirement, but with a new training run on genuine manufacturing debris, and an inference method that splits screenshot images into multiple smaller sections for YOLO to detect particles on, the issues with small particles would likely be resolved.

Failure Mode and Effect Analysis

Full FMEA Chart:
https://docs.google.com/spreadsheets/d/1oW0sy-NdEdZSj3KsKEz6RjZHcGIsb9MpDl0_YOcfLVA/edit?usp=sharing

To summarize the FMEA chart, the major potential modes of failure mainly involve issues with image analysis accuracy, image quality, and stage controller movement. Incorrect particle sizing or missed detections by the image analysis software can lead to inaccurate measurements and a higher rate of false positives, which can be mitigated by performing a quality assessment at the start of each shift. Blurry or unfocused images can contribute to these issues, which emphasizes the need for routine station cleaning and checks. Additionally, if the stage controller does not move the intended distance, it may result in incomplete filter scanning, which can be prevented through proper calibration and following the qualification test procedure.

Potential Failure Modes	Potential Failure Effects	Actions Recommended
Image analysis software incorrectly sizes particle	Generates inaccurate measurement, increases false negative detection	Perform quality assessment before each shift begins
Image analysis fails to identify particle	Generates inaccurate particle count, increases false negative detection	Perform quality assessment before each shift begins
Acquired image is too unfocused	Image will be harder to analyze leading to Failure Modes 1 and 2	Routine check, clean station before morning shift begins
Stage controller does not move intended distance	Filter cannot be fully scanned	Calibration, follow the qualification test procedure

Table C: Major Potential Failures and Mitigation

Minor potential failure modes mainly relate to misalignment and improper setup of the imaging system. Incorrect filter placement or misaligned stage controller planes can interfere with accurate image capture, potentially missing parts of the filter. Similarly, a dirty or misaligned microscope lens may produce blurry images that hinder analysis, but these issues are easier to detect early in the metal particulate test. These minor issues can be effectively addressed through routine check-ups, calibration, and strict adherence to the qualification test procedure.

Potential Failure Modes	Potential Failure Effects	Actions Recommended
Filter is placed incorrectly	Can't run image gather script accurately because filter is not in correct position	Calibration, follow the qualification test procedure
Stage controller planes are not aligned	Some edges of the filter will not be captured	Calibration, follow the qualification test procedure
Microscope lens is dirty or	Images will be blurry and hard	Routine check-up, follow the

misaligned	to analyze	qualification test procedure

Table D: Minor Potential Failures and Mitigation

Lessons Learned

During the manufacturing process of the 3D-printed knobs, the first iteration of them were printed using PLA filament on a cheaper printer with looser tolerances. This made the fit between the knob and the stepper motor imprecise as the knobs' fitting was too tight and caused the end of the knobs to bulge outwards. The 3D printer and material was the cause for this issue as they are not made to print smaller components. The solution to that problem was to utilize a 3D printer that was designed to produce smaller and more complex parts. An SLS printer was used with a material called Nylon 12 powder to make a part that has tighter tolerances, allowing for a perfect fit between the knobs and stepper motors. When 3D printing, it is important to know what is the tolerance of the 3D printer and also select the type of filament that suits the needs the best.

One issue in this project is to set up the circuit and get everything to work together. Some difficulties that were encountered consist of getting one stepper motor to move as designed. There was some stagnation of the 5V stepper motors where the motor could not get the stage control to move efficiently. Originally 5V stepper motors were used for the circuits, but it did not have enough torque to move the stage controller efficiently as stagnations of the motors were observed. As a solution, 5V stepper motors were replaced by 12V stepper motors, and the motors were supplied with a stronger power supply of 12V DC power supply. Having more power and more torque to the stepper motors definitely were the first step to getting the prototype to work.

Another set of issues occurred when calibrating the distance per one step of the stepper motor. Originally, we used the help of ChatGPT to calculate the number of steps on the stepper motors for it to move 2X2 squares on the filter membrane while looking through the microscope. The distance did not come out to be as accurate as it was designed to be, all the movements were misaligned. Therefore, a manual recalibration was needed; multiple tests were run concurrently to get the stepper motors to move the stage the exact distance that it was supposed to move. Some simpler methods could have been implemented to make the calibration fast and efficient, but this manual recalibration was the only method at the time.

Establishing serial communication between Python and Arduino also was an issue. The python script and Arduino serial communication code were also correct, but for some reason, Arduino was not able to follow the command sent from the Python script. The solution for this very simple problem was to open the serial monitor in Arduino first to check if the serial communication is set and Arduino is ready to receive command from Python. After the "Ready" sign is shown in the serial monitor, close the serial monitor tab, then run the Python script, python script would not work if the serial monitor is not closed in Arduino because only one software program is able to access a serial port at a time.

Implementing the YOLO AI model to the python script also was a problem for quite a bit. The AI could be placed in the script, but it could not be loaded for particle detection. The problem was caused by the fact that the directory that was put in the Python script to load YOLO was incorrect and the AI folder was not inside the project folder where it contains the functioning python script. Getting the directory right is very significant in this process, otherwise the Python script will not be able to locate and load the AI model.

The AI model does not have that high of an accuracy yet when detecting the titanium particles as false detections and missed particles were observed. One of the reasons for the missing particles was simply because YOLO needed more training data to recognize the titanium particles since it was not trained with a variety of particle sizes and shapes. Another big issue was the false detection when the blue plastic section of the stage control top was captured. When the microscope camera auto exposure feature was put on the stage top as the stage moves, the color of the blue area is darker than the white filter membrane, which caused the auto exposure to adjust the overall brightness of the screen. The blue section would then become very bright, causing the area to simulate the shininess of titanium particles which resulted in a false detection. The first solution was to use a method called blue region masking, OpenCV was used to mask out the blue section of the stage top so YOLO would only detect objects inside the filter membrane. However that is not a very robust solution as that would lead to some missed particle detection. So the best solution moving forward would be to train the YOLO model to disregard the blue section of the stage top or to change the stage top to a white color so there won't be any auto exposure issue.

Hardware Training Manual

Assemble Physical Components:

1. Describe Required materials: motor holder, stage controller, stage top, screws and bolt, stepper motor, knobs

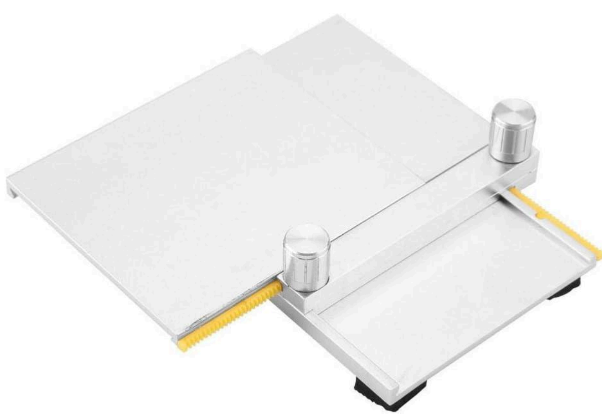


Figure 8. Stage control

2. Stage controller - glue stage top

Figure 9. Stage Top

3. Stage controller - remove original knobs and put in 3d printed ones

Figure 10. 3D printed knobs

4. Stepper motor and motor holder - screw down motors to motor holder and place holder on stage

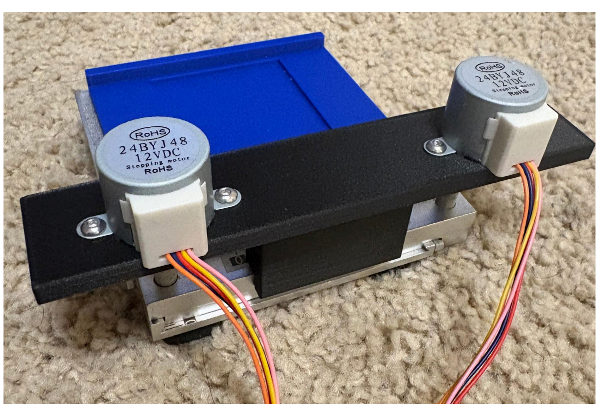


Figure 11. Stepper motor mounted on stage control

Assembling Circuitry:
First of all, to set up the stage controller these materials will be included: 12V Bipolar stepper motor module, 12V DC power supply, female power supply terminal screws terminal, stage control, and Motor holder.

The first step is to connect each of the 12V stepper motors to its driver board with male/female wires. Attach both motors to the motor holder and screw them to maintain stability. Now connect the driver board to the Arduino controller board (input 4 on each stepper motor driver board will be wired to the lower digital pins. For instance, IN4 will be connected to digital pin 8, and IN3 will be connected to digital pin 9. Having the input pins and digital pins set up incorrectly will result in a movement malfunction.

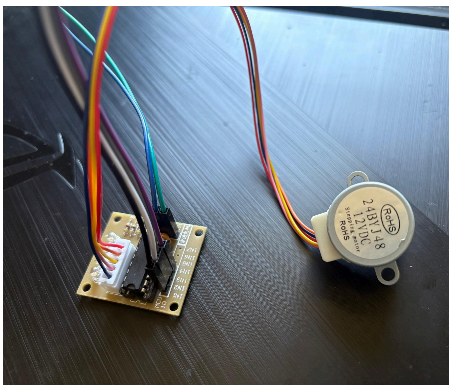


Figure 12.
Stepper motor + driver board

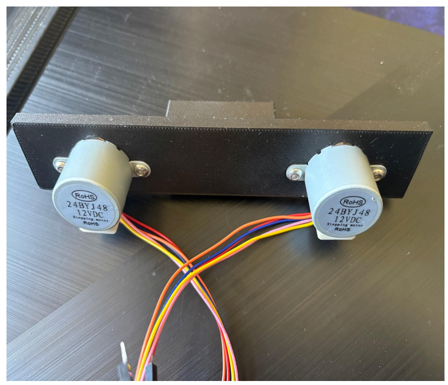


Figure 13.
Stepper motors connected to a holder

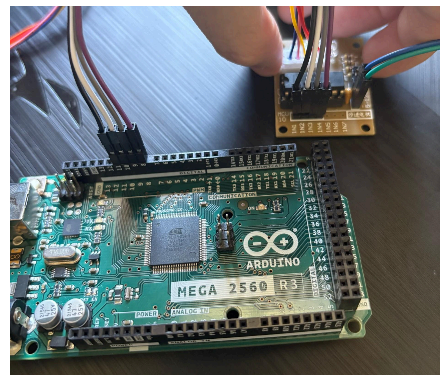


Figure 14.
Arduino 2560

The motor driver will also be connected to a breadboard to receive power from DC power supply. Make sure that the input pins wires connected to the digital pins of both motors do not touch each other, or else it can cause an electrical malfunction and signal conflict (movement of stepper motor will be impaired).

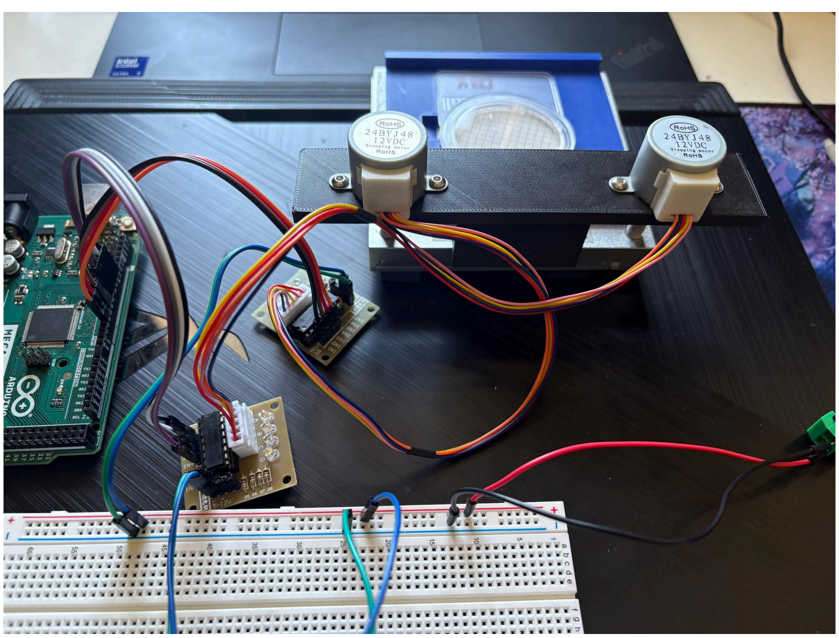


Figure 15. Stage controller connected to breadboard to receive power

To check whether the setup is correct or not, expect to see two LED lights on both of the stepper motor drivers lit up when the code has been run. That will be an indication that the motors are functioning properly. In some cases that only one LED is lit up, it is very easy to tell since the motor would not move as efficiently as it should (Stagnation of the motor can be observed), check the wires to see if there's any misplacement between the input pins on the motor drive and the digital pins.

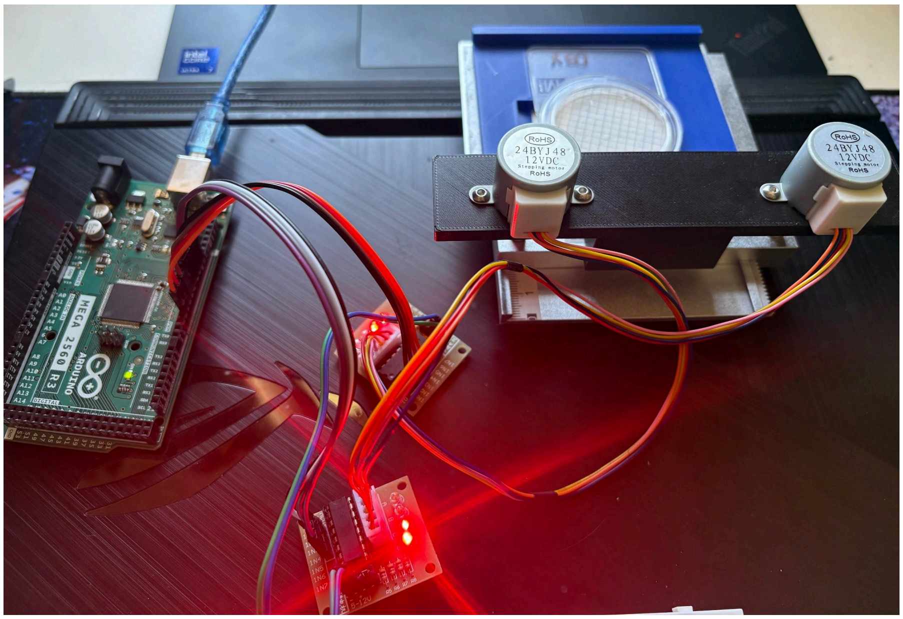


Figure 16. Stage controller performance testing with LEDs

After finishing all the steps, the stage controller can now be placed under the microscope. Plug the microscope into a power outlet, then turn on the microscope light. You would want to control the live video through a software program called AmLite. Adjusting the magnification to the point where the microscope camera captures 3X2 squares of the filter area. Now the code is ready to be run.

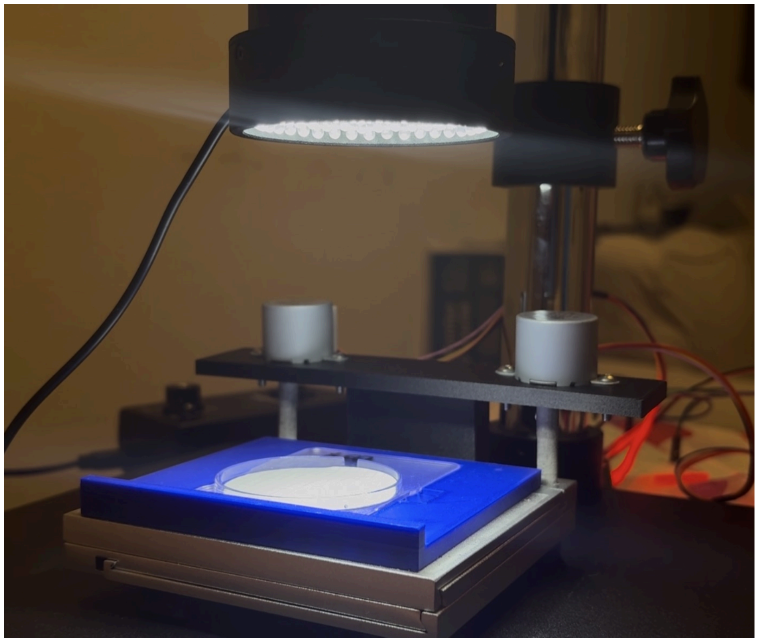


Figure 17. Final look of the prototype

User Software Interface Manual

As for how the code will function, the Arduino code will be used to control all the function of the stage movement. When Arduino code is run, there will not be any movement, since it only contains function code to follow commands from Python. The serial communication is established with Python so that the stage controller would only move when it receives a string command from python.

Python software would send string commands to Arduino over serial communication. After each movement of the stage controller is done, it will send a message "done", in which the python would then capture a screenshot and save the image into images file, then proceed to the next step. As for right now, the stage controller would start from one position, covering all the area of the filter while capturing images between each pause of the movement. The stage would then move back to its original position.

To change the distance of the stepper motor, an easy adjustment is to change the number of steps on the stepper motor in each movement in the Arduino code.

To get the code to run, we would have to run Arduino code first, open the serial monitor to check for "Ready". If the "Ready" is not visible, close the serial monitor and reopen it. Once there is a ready sign, it means the Arduino code has reset the serial connection and is now ready to receive commands from Python. Now that the Arduino is ready, close the serial monitor and run python code. The reason why the Arduino serial monitor has to be closed first is because only one software program can access the serial port at a time.

To implement the YOLO ai model to the python, the YOLO file should be placed where the python script can access, which is inside the project folder where it contains the codes. In that case the python code will be able to direct a path to load the YOLO model. If the directory of the YOLO folder is incorrect, the model will not be able to access the model and if the YOLO model folder is not being placed inside the project folder, it also cannot be loaded.

Once the code is run and all the images are captured, annotated images with particle detection via drawing the bounding boxes will be present along with the measurements. All the annotated images will be saved into a folder.

How is the device to be used by the user?

The automated particle detection system is designed to support the users in evaluating filter membranes during the testing process of a handpiece medical device. It integrates a motorized stage controller, a microscope camera, image acquisition software (AmLite), and an image processing artificial intelligence model called YOLO for object detection and OpenCV for particle measurement.

Once the system is fully set up and the stage controller is aligned under the microscope, the users initiate the test by running the Arduino code followed by the Python code. The stage controllers move incrementally across the filter membrane as commanded. After each movement, the microscope captures a high which is stored in a designated folder for later processing.

The users are not responsible for any mechanical input during the time. Instead their primary task is to observe the automated process and verify that the system performs correctly. These monitorings include:
Confirming image quality: Ensuring each screenshot captures the filter area clearly with proper contrast and focus. Overly exposure can result in overly bright or saturated images which can imitate the shininess of titanium particles under shining microscope light, this would result in false detections.
Monitoring serial communication: Check for “Ready” in the Arduino serial monitor before closing it and run python code. Python code will not work if the serial monitor tab is not closed in Arduino since only one software can access the port at a time.

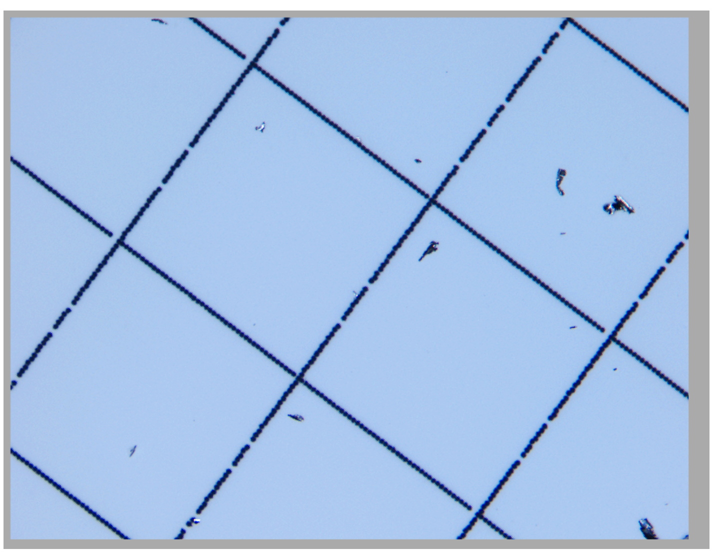


Figure 18. Filter membrane image

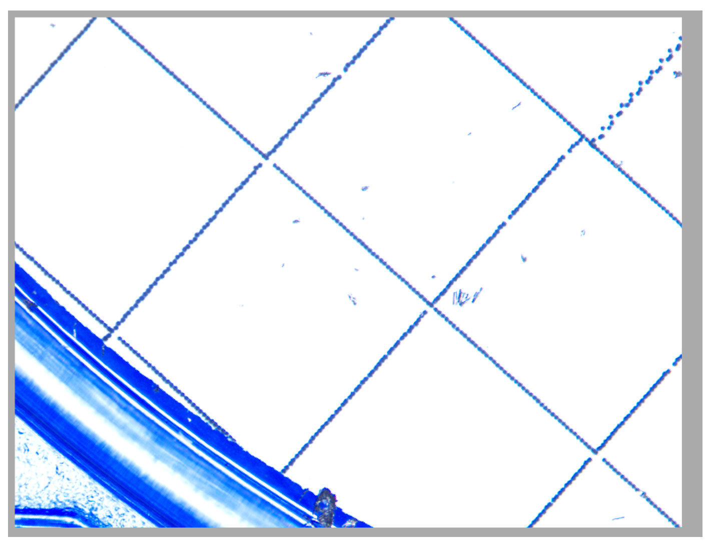


Figure 19. over exposure of auto exposure

How users can spot errors:

To ensure accurate detection and optimize performance during the prototype phase, users should be wary of the following errors and malfunctions:

Overexposure of the microscope image:

If the live image displayed in the software AmLite is too bright, look to adjust the position of the auto exposure to the middle of the filter membrane. If the auto exposure is on the plastic section of the stage top, the blue color is darker than that of the filter membrane, which is white, the camera auto exposure system will try to adjust the overall scene brightness by increasing the brightness. And when this occurs, YOLO will misidentify the plastic section as titanium particles.

Unusual bounding box:

When YOLO detects non-particle features (e.g., part of the blue plastic housing or grid lines), the bounding boxes may appear around those features. These can be spotted easily as they deviate in shape or size from typical debris.

Inconsistent particle counts:

When all the images are captured and being loaded for YOLO to analyze, if all the particles are not detected, then that would indicate detection errors and should be reported to the software team for them to provide YOLO with more training data.

Stagnation in the stage movement:

Any signs of stage or stepper motor stagnation may suggest wiring issues between the input pins and the digital pins. One other issue it can be is the overheating of the stepper motor. If the stepper motors become too hot, it can become stagnant or experience issues like losing steps, which means it won't move to its designed position accurately. Identifying these problems early on will help the engineering team replace the damaged parts with new functioning components.

Functional Trials

Video link:
https://drive.google.com/file/d/1XT6JQiSE6myEB3ONZPAdtSxrjxl8O6we/view?resourcekey

Other than problems identifying the smallest titanium particles, our device functions as intended. The device starts up, moves the stage, captures images, performs object detection inference, and outputs results successfully.

Future Directions

If the team was given unlimited resources, a $1 million R&D budget and another year to work on this project, the design of the automatic stage controller will be vastly different from the current design. The current design was limited by the time frame that the team was given, which was only six months. A new stage controller will be made entirely from scratch, and it will be controlled using CNC code instead of Arduino. The circuitry can also be changed to a printed PCB, which would be more compact and reduce space. Making a stage controller from scratch eliminates the inaccuracies in movement associated with using a premade mechanical stage controller. In addition, the extra money and resources will allow the team to explore physical methods to increase the visibility of the titanium particles by enhancing the contrast between the particles and the background. Finally, the team would allocate more resources in the recognition of other extraneous objects besides titanium particles which could potentially be dangerous to the patients. For the software side, the extra time will allow for new data sets on filters that portray real failed tests from Alcon with the extra resources. The data set that was used to train the image analysis software used only filters that passed the examination or simulated failed slides that do not accurately represent a real failed filter. In addition, the extra money would allow the team to purchase a higher quality microscope with a better resolution and smaller FOV to take higher quality images, which would increase the accuracy of detection of the image analysis software. These changes would be more optimal for high volume manufacturing of the system as most of the components can be made within Alcon instead of requiring outside providers for the current design’s components.

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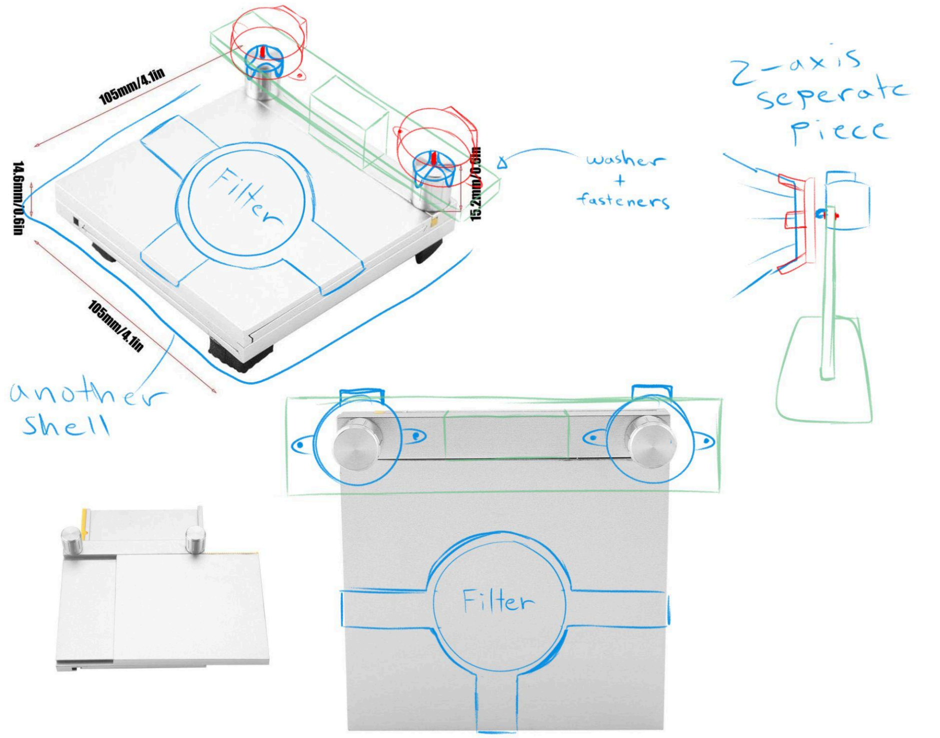
[18] Alcon, “Manufacturing Operating Procedure for Digital Measurement Microscope,” internal procedure V-QMS-0089426, Alcon, Fort Worth, TX, USA, n.d.

Appendices

Appendix 1: Design Documentation

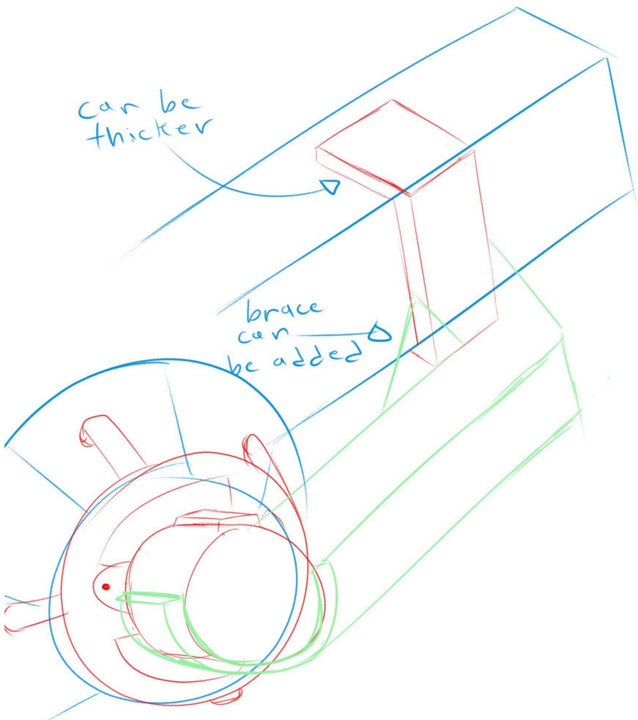
Initial Sketch:



Explanation:

- Two servo motors will control each of the axes on this prebuilt stage controller
- A gripper will be made in Solidworks and attach to the end of the servo motor
- To balance the motors, a platform is made to have the motors attach to the block with washers and fasteners
- Platform will extend past the back to prevent the platform from losing balance
- A shell will put the filter in a fixed position
- A shell will put the stage controller in a fixed position
- The z-axis controller will be a separate piece but controlled by the same circuitry as the main piece

Z-Axis Redesign Sketch:



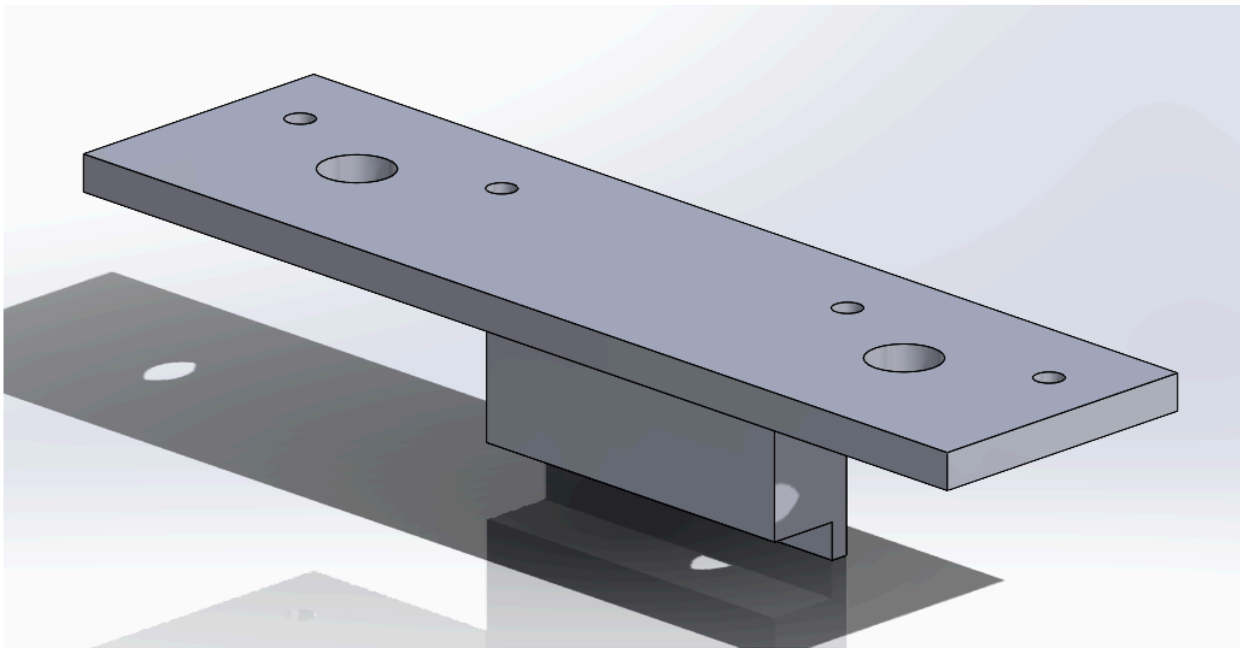
Explanation:

- Servo needs to be redesigned because the focus moves with the microscope
- An attachment is made to the brace of the microscope moving mechanism
- The base of the attachment can be made thicker for better balance
- Brace can be added for better balance

Design Sketches Against Pugh Matrix:
https://docs.google.com/spreadsheets/d/1iGcm1S9z8DWJZS40kyCQZ9n-N9m_iC955W4UqYiYVp8/edit?usp=sharing

Platform Solidworks Design:

Model:



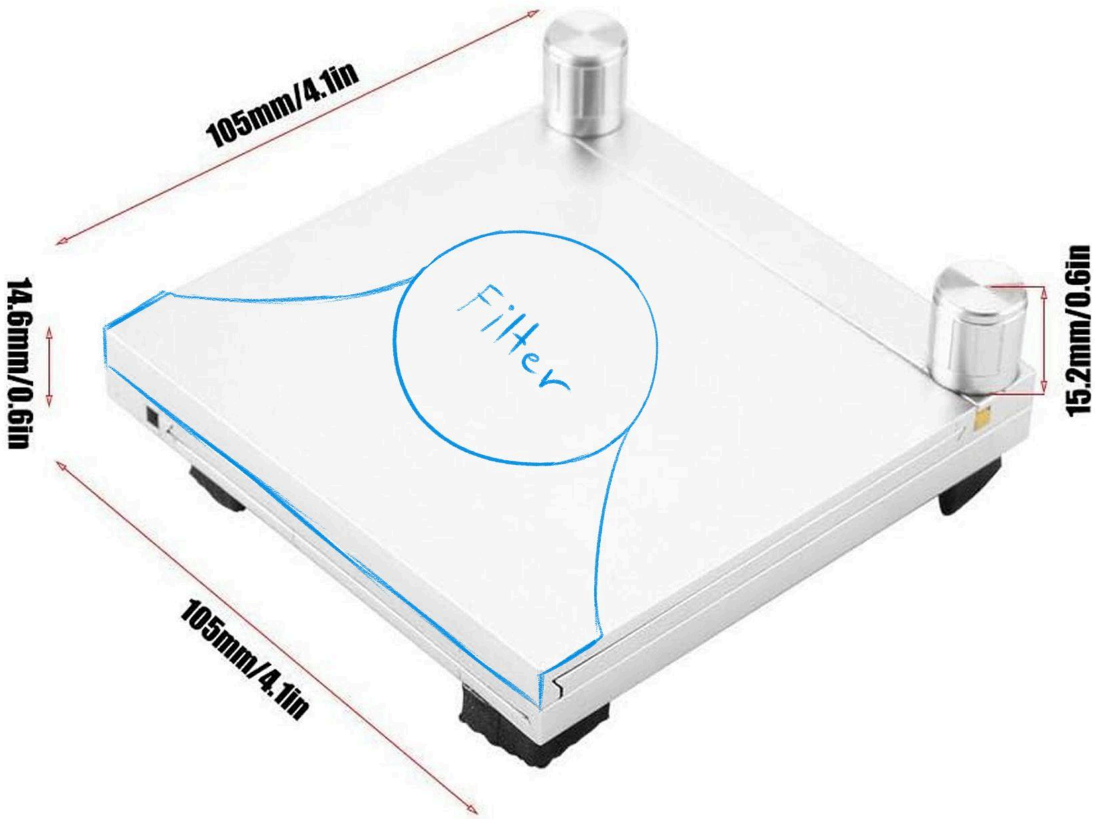
Explanation:

- Thinner top platform
- Can be glued to stage controller
- Extra braces can be added for extra stability

Plate Shell Solidworks Design:

Measurements:
- Wideness of 105 mm
- Thickness of 0.75 cm

Sketch:



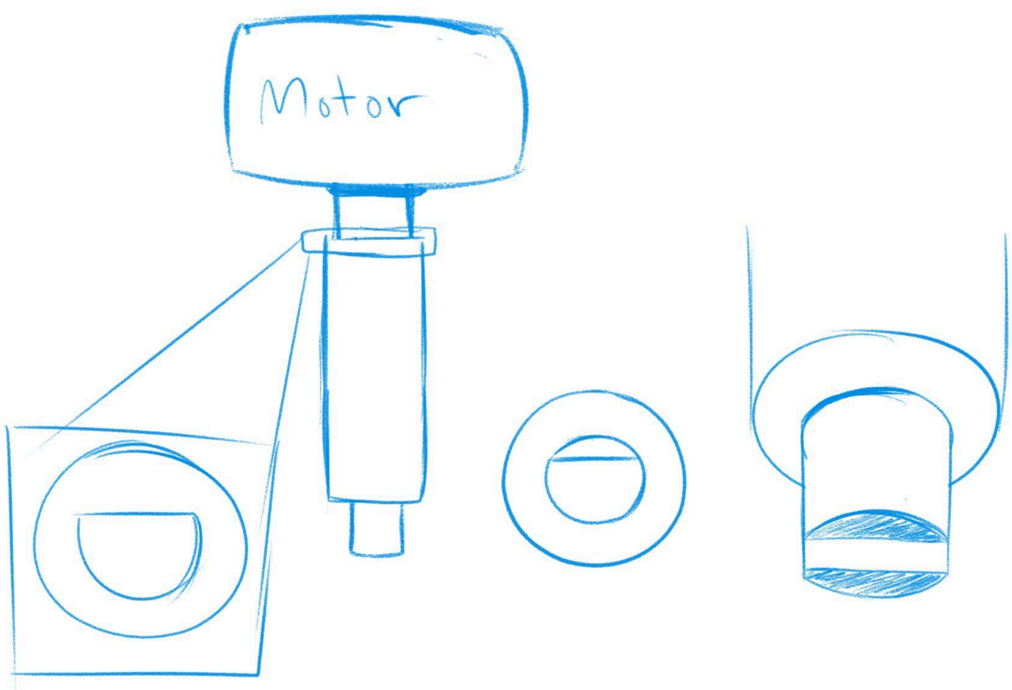
Explanation:
- Will be glued on
- Curves not needed, can be straight lines
- The filter has to be in the middle (design the square first then make the shell)

Knob Grippers Solidworks Design:

Measurements:

- Knob is 16.5 mm
- Knob screw is 12.5 mm
- Knob screw diameter is 4.3 mm with 3.65 mm cut off
- Knob screw diameter 2 is 5.9 mm

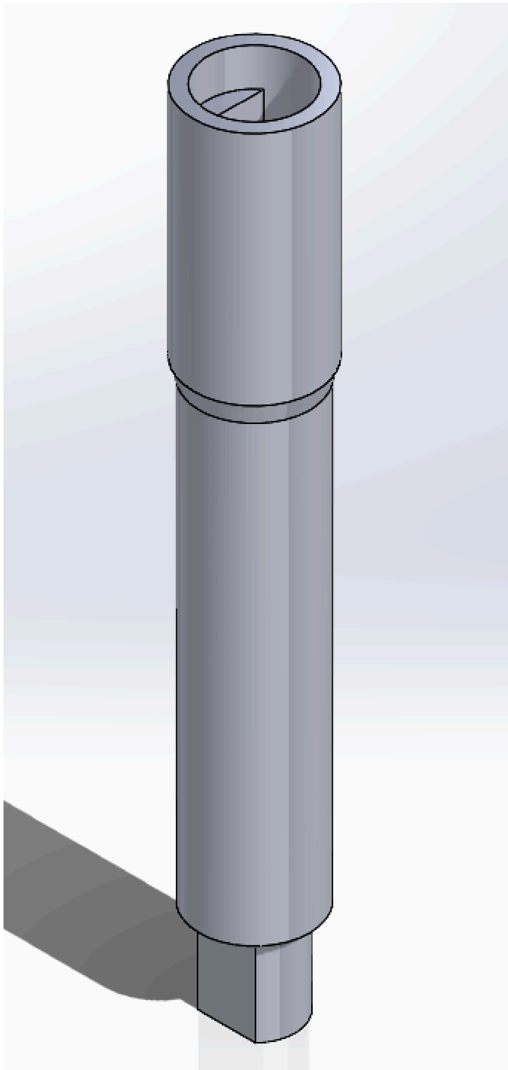
Sketch:



Explanation:

- One piece that will replace the original knobs and connect it directly to the stepper motor

Model:



Explanation:

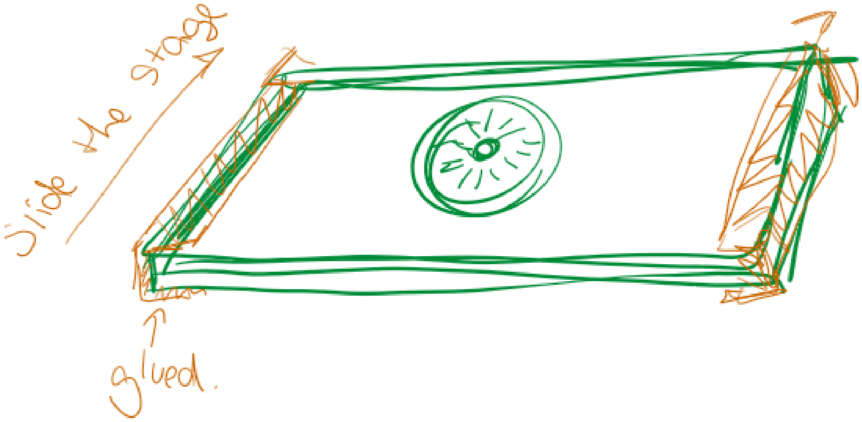
- Thicker top side for extra stability
- One solid piece that connects the stage controller and stepper motor
- The holes are designed with the Arduino stepper motors in mind

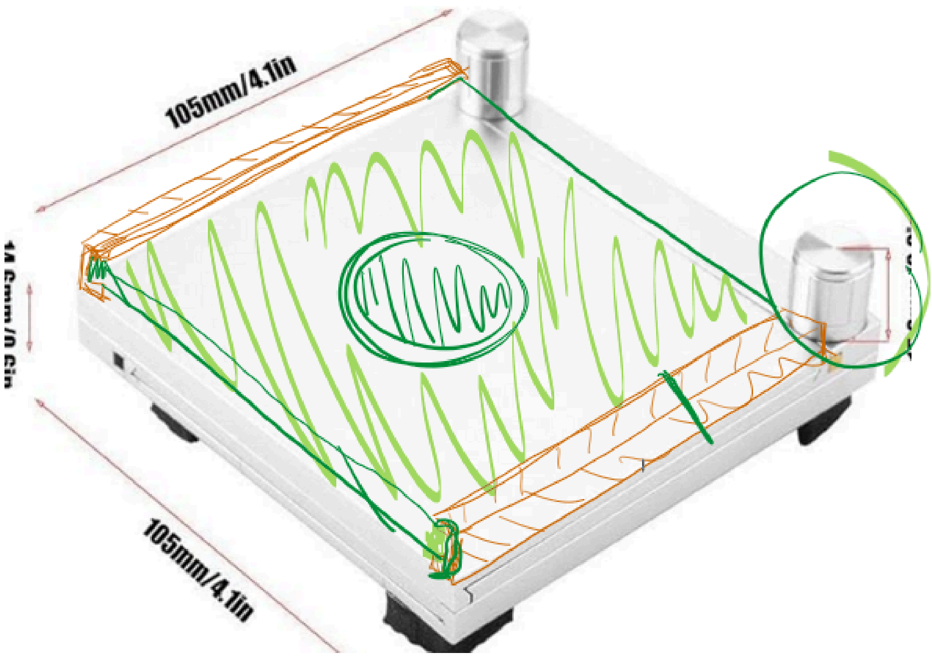
Stage Controller Shell Design:

Measurements:

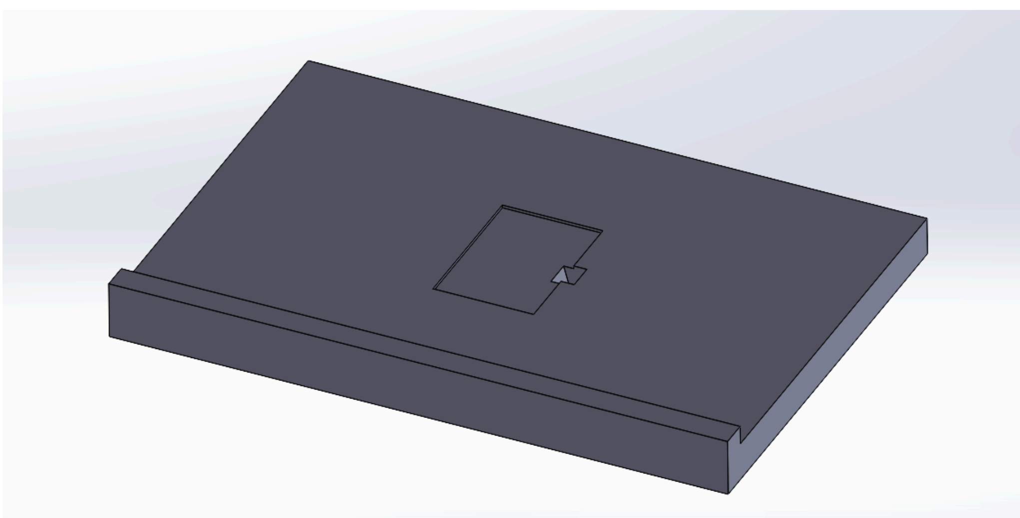
- Base is 32 cm x 22 cm x 2.4-2.5 cm
- Stage controller should be placed in the middle of this, so design a shell with this in mind
- Remember that the filter plate moves out, so the shell cannot interfere with the plate movement
- Since the stage is lifted by the feed pads, try and build something under that

Stage Top Idea



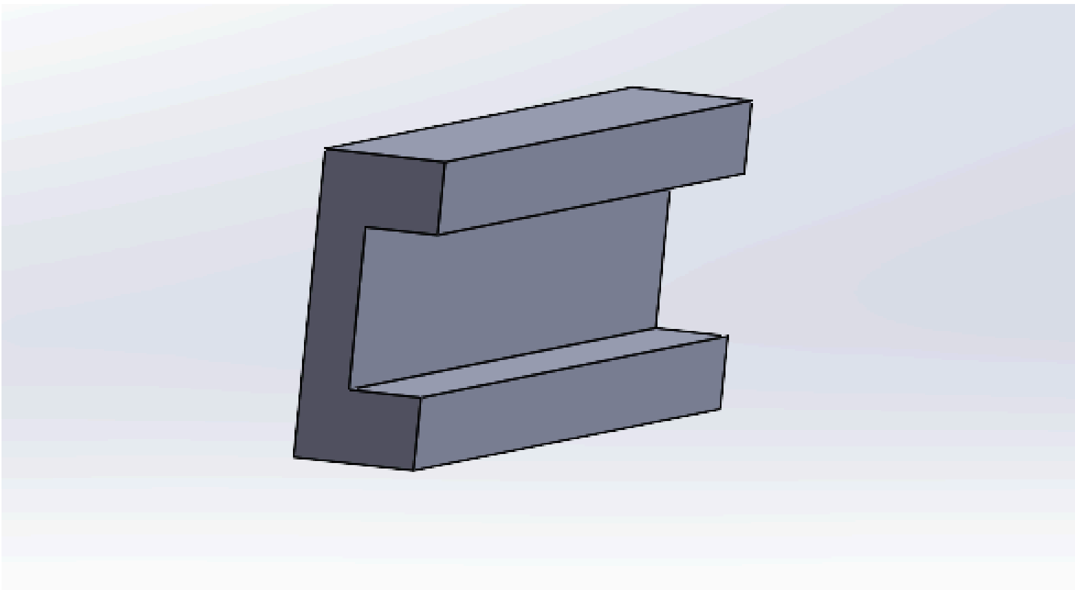


Stage Top Design



Details:

- Stage contains a slot for the entire filter case as it is not removed from the case while it's being processed for particle counting
- There is also an extra space right next to the case slot for easy removal once the particle counting with the microscope is done
- The measurement for this design is the same as the already bought stage controller



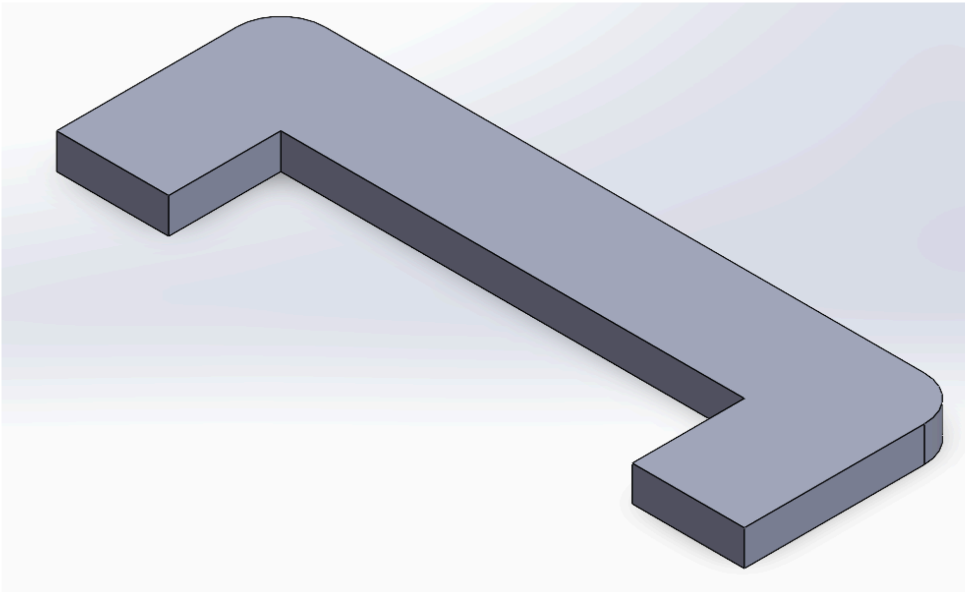
Details:

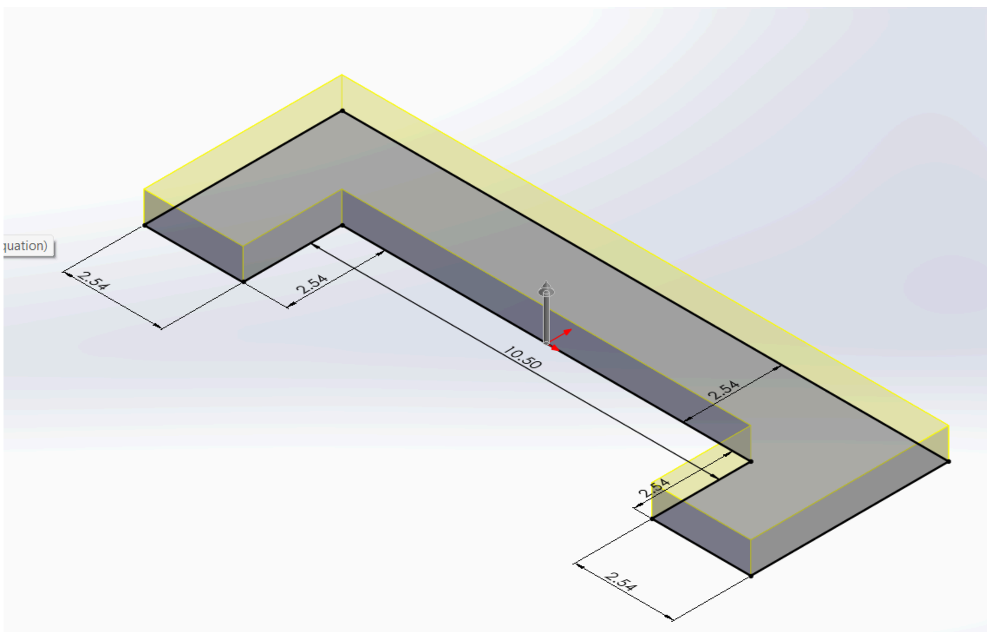
- We'll print two of these
- The design is the stage top track, they will be attached to the bought stage controller so that the stage top can slide in and out whenever the processing is done

Note:

- If any of these measurements obstructs the mobility of other parts, we can always file them and re-measure for final prototype

Idea to align stage controller to microscope
Purpose: To ensure the stage controller is always placed in the same position under the microscope after removal and reattachment





(extrude 0.9 cm)

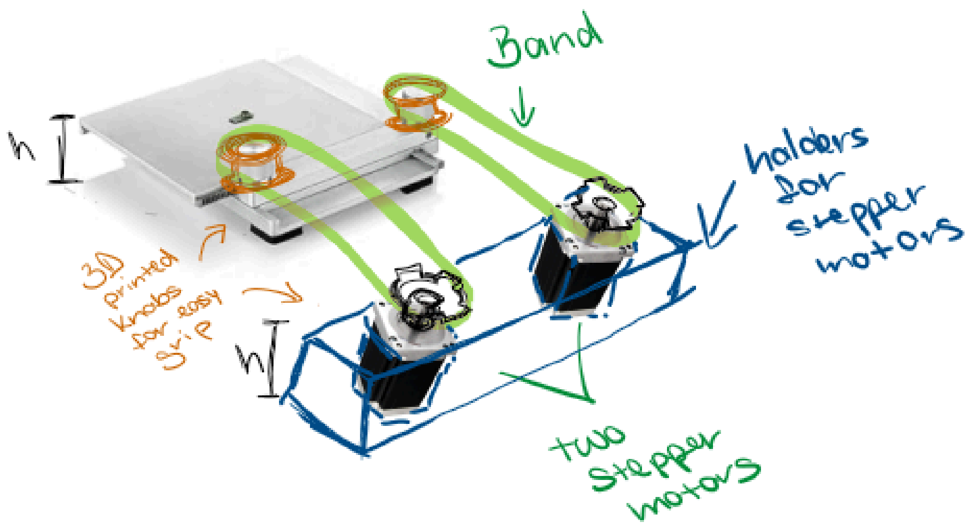
What to order:

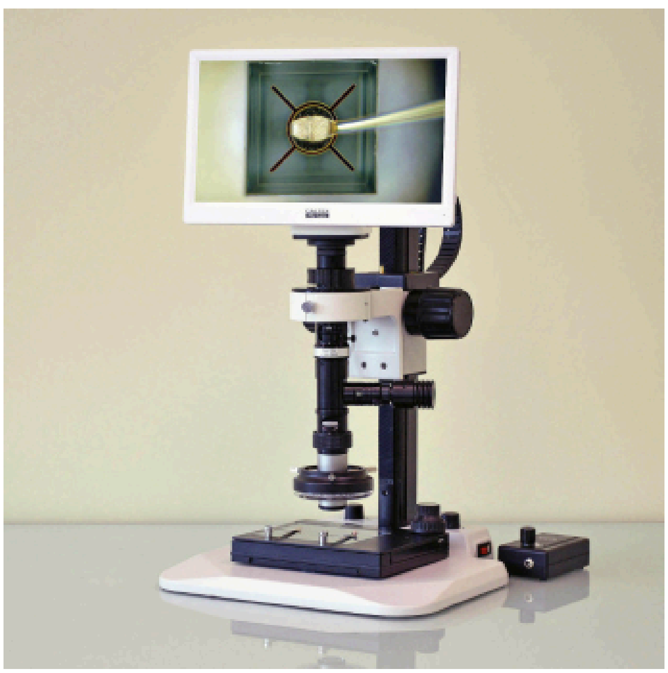
1. 3D print order for this
2. Some adhesive
    a. Double sided tape
    b. Adhesive Foam (take into account tolerance)

Additional Idea: Laser positioning

3. Laser pointer
4. Ziptie
5. Sharpie

Item	Link
Glue	Reuse from plate shell design
Laser pointer	https://www.amazon.com/Presentation-Clicker-Wireless-Presenter-Remote/dp/B0D8W6CYF2/ref=sr_1_20_sspa?crid=2SDOSVV00DCRI&dib=eyJ2IjoiMSJ9.a0DOUbrPHrXY7w7IFI-v6PGEYxA9zJcIWhVc1sS0eNaRB2_6tCGusrxrvfLxj5xIdBk3EVq7g2CJof8fgri_Yx6MZqwvBwOLlqPfr9Ea7I-Zh5mWtqaHTvQNdQnxUfEpHbBHi0-IwGWWAAJ3H0CercWPv04nqFG_qDp0RZxvAg8A8dqpPySQLlSSf3EL_bjeDvnYDX_e6r95VXHOA_kyJ6y0FvyVNPSTmBT6wjEtBiOo-9BrISDRWvRn1TN9CTXLs0Qi4drhOe9kpeHzFctbGrXrcMSwqBRuyHNuKUACg28.vNRhs7k-GifTSNW0v_J3jpZRaLBDEx6wWgxcEH4Qksk&dib_tag=se&keywords=laser%2Bpointer%2Brechargeable&qid=1739906313&sprefix=laser%2Bpointer%2Brech%2Caps%2C222&sr=8-20-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9tdGY&th=1
zipties	Hopefully UCI has some lying around
sharpie	Any dot item that aligns with laser for repositioning


Band
h
holders for stepper motors
3D printed knobs for easy grip
h
two stepper motors




Knob
gripper
holder

Project Plan

Student Learning Outcome/Project Constraint	We considered this in our project (Yes/No?)	It was Relevant in our project (Yes/No?)	Location in Plan Where this is Addressed
Apply principles of engineering	Yes	Yes	Throughout, design, manufacturing, research, validation
Apply principles of science	Yes	Yes	Throughout, design, manufacturing, research, validation
Apply principles of mathematics	Yes	Yes	Throughout, design, testing, software development, validation
Constraint: Design with public health consideration	Yes	Yes	Throughout, research, design, manufacturing, validation
Constraint: Design with safety and welfare consideration	Yes	Yes	Research, design, manufacturing, validation
Constraint: Design with global consideration	No	No	N/A
Constraint: Design with social and cultural consideration	No	No	N/A
Constraint: Design with environmental consideration	No	No	N/A
Constraint: Design with economic consideration	Yes	Yes	Throughout, manufacturing, validation

Constraint: Design with manufacturability consideration	Yes	Yes	Design, manufacturing, validation
Constraint: Design with ergonomic consideration	Yes	Yes	Design, manufacturing, validation, software development
Constraint: Used ethical considerations to inform solution in global context	Yes	Yes	Design, software development

Constraint: Used ethical considerations to inform solution in economic	Yes	Yes	Design, manufacturing, validation, software development
Constraint: Used ethical considerations to inform solution in environmental	Yes	Yes	Manufacturing
Constraint: Used ethical considerations to inform solution in societal context	Yes	Yes	Design, validation
Team members who provided leadership	Yes	Yes	Matthew Thai Luong, Yolanda Aolani Gomez, Tandol Mathew Jerthsattayan…, Brian Ngo, Landon Stein, Isaac Robert Hoyt
Team members who provided collaborativeand inclusive environment	Yes	Yes	Matthew Thai Luong, Yolanda Aolani Gomez, Tandol Mathew Jerthsattayan…, Brian Ngo, Landon Stein, Isaac Robert Hoyt

Team members and activities who established goals, planned tasks, and meet objectives	Yes	Yes	Matthew Thai Luong, Yolanda Aolani Gomez, Tandol Mathew Jerthsattayan…, Brian Ngo, Landon Stein, Isaac Robert Hoyt
Used appropriate Standards and experimentation	Yes	Yes	Design, manufacturing, validation, testing
Analyzed and interpreted data to draw conclusions	Yes	Yes	Software development, manufacturing, testing
Acquired and applied new knowledge as needed for your project	Yes	Yes	Throughout
Applied appropriate engineering standards to your design	Yes	Yes	Design, validation
Applied appropriate FDA standards to your design	Yes	Yes	Design, research, validation